The Information Costs of Climate Misinformation:
Evidence from Global Financial Data

Abstract

Climate misinformation can alter the information environment in which investors and liquidity providers trade. This paper examines that relationship in a panel of 204 listed climate-transition securities, using sector-specific exposure constructed from 217,954 English-language Facebook climate posts classified against nine energy sub-industries. Greater exposure is positively associated with quoted bid–ask spreads under monthly and local market-day controls. On a common sample of 486,655 security-days, the market-day estimate is +0.0129 percentage points per index point ($p = 0.033$ under sector/date clustering), a pattern consistent with information-verification costs. The association changes sign and is imprecise after sector-month conditions are absorbed; with prior-session financial controls, the estimate is −0.0014 ($p = 0.533$). Timing and alternative-outcome checks do not establish an additional robust response. The findings connect the political production of climate misinformation to the trading environment of energy equities while locating the positive spread association in shared sector conditions. The contribution is a sector-sensitive measure and a transparent account of where the liquidity association is observed, rather than a causal estimate of misinformation costs.

Keywords: Climate Misinformation, Information Asymmetry, Market Microstructure, Environmental Communication, Politics and Governance, Corporate Responsibility.

1 Introduction

The information landscape surrounding climate change has been systematically distorted by misinformation campaigns (Cobb, 2025). Markets aggregate information, but they do so at a price. Climate misinformation is produced for political ends—funded by fossil-fuel interests, amplified by ideologically motivated actors, and timed to political events (Farrell, 2016; Oreskes and Conway, 2010; Cinelli, De Francisci Morales, Galeazzi, Quattrociocchi, and Starnini, 2021)—yet market-microstructure theory gives two opposing predictions for its effect on trading costs. If misinformation increases the advantage of sophisticated traders who can distinguish credible signals from false ones, liquidity providers should widen bid–ask spreads to protect themselves. If misinformation instead attracts uninformed retail trading, the perceived share of informed order flow may fall and spreads may narrow even as the market's capacity to absorb informed trading deteriorates. Which pattern prevails is the central question of this paper.

This question connects two related literatures. Political-behavior research shows that misinformation changes beliefs, trust, and collective action (Flynn, Nyhan, and Reifler, 2017; Bursztyn, Rao, Roth, and Yanagizawa-Drott, 2020), while climate-finance research shows that climate news, disclosure, and policy uncertainty are priced in financial markets (Engle, Giglio, Kelly, Lee, and Stroebel, 2020; Bolton and Kacperczyk, 2021; Pankratz, Bauer, and Derwall, 2023). Neither literature traces the daily flow of politically produced climate misinformation into the mechanism that sets trading costs. Market microstructure provides that link because the quoted bid–ask spread is an observable price of information asymmetry and a direct measure of market liquidity (Glosten and Milgrom, 1985; Easley and O'Hara, 1992).

The theory distinguishes two competing channels. Under adverse selection, false claims about climate science, carbon policy, and the viability of low-carbon technologies make the implications for fossil-fuel and renewable firms harder to assess. Traders able to verify those claims gain an informational advantage over those who cannot. Liquidity providers then face greater expected losses when trading climate-transition equities with better-informed counterparties and may widen spreads. Under liquidity fragility, viral climate-policy and energy narratives instead draw noise traders into the affected sub-industries, potentially narrowing spreads as the uninformed share rises. H1 tests the narrowing-spread prediction. Section 3 develops the mechanisms and distinguishes their theoretical implications from the information available in daily quote data.

The empirical analysis starts with 726,890 security-day observations on 204 listed climate-transition securities, representing 198 distinct company names, from January 2012 to December 2025. Financial

estimation is restricted to the content index's observed window, January 2012–October 2025. The primary outcome is the percent quoted bid–ask spread. The exposure measure is constructed by classifying 217,954 English-language Facebook climate posts from the Meta Content Library against nine TRBC energy sub-industries. Each security inherits its sub-industry's exposure on each date, producing within-day cross-sectional variation while preserving the political and economic content of the posts. Correctly mapped listing-market effects, sector-month controls and prior-session characteristics assess how locally the spread association can be identified.

Greater sector-specific misinformation exposure is positively associated with quoted spreads in the monthly and local market-day specifications. The latter coefficient is $+0.0129$ ($p = 0.033$ under sector/date clustering), supporting the view that the climate-information environment and trading costs move together. This positive association is not stable after sector-month conditions are absorbed: on the same sample the coefficient becomes -0.0033 ($p = 0.117$), and the prior-session-control specification yields -0.0014 ($p = 0.533$). The evidence therefore identifies a positive association in the broader sector information environment while placing a clear boundary on a daily misinformation-specific interpretation. The null estimates are part of that finding, rather than evidence that climate misinformation is financially irrelevant.

The paper makes three contributions. First, it extends market-microstructure theory to politically produced climate misinformation and makes the competing adverse-selection and participation predictions explicit. Second, it provides a high-frequency sector-sensitive exposure measure that maps political misinformation to the sub-industries whose values depend on the contested information. Third, it shows that a positive association remains after local market-day conditions are absorbed, while sector-month controls materially change the identifying comparison. Together, these contributions connect the political production of misinformation to the liquidity of climate-transition equities and clarify which part of the relationship the observational evidence supports.

The rest of the paper proceeds as follows. Section 2 reviews research on information and market liquidity and connects it to climate misinformation in financial markets. Section 3 develops the competing adverse-selection and liquidity-fragility mechanisms and states the hypothesis formally. Section 4 introduces the firm-day financial panel and misinformation corpus. Section 5 separates the market-liquidity design from the benchmark risk-and-return designs and sets out the identification and inference strategy. Section 6 reports the main estimates, relevance checks, and robustness comparisons. Section 7 interprets the findings, and Section 8 concludes.

2 Literature Review

2.1 Information and Market Liquidity

Market liquidity depends on the distribution and credibility of information. When some traders possess information that others lack, liquidity providers face the risk of trading at stale prices with better-informed counterparties (Akerlof, 1970; Glosten and Milgrom, 1985; Easley and O'Hara, 1992; Kyle, 1985). They recover the expected loss through the bid–ask spread, while price impact records how strongly order flow moves prices. These measures make the trading-cost implications of information asymmetry observable. Costly information prevents markets from becoming fully efficient (Grossman and Stiglitz, 1980a; J. E. Stiglitz, 2000), and liquidity providers revise quotes as the perceived composition of informed and uninformed order flow changes (Hasbrouck, 1991; Easley and O'Hara, 1996). Wider spreads indicate lower market liquidity and can pass through to firms' costs of capital (Tse, 2024; Aliyev, Aquilina, Razayev, and Zhu, 2025; Aquilina, Ibikunle, Rzayev, and Wang, 2025).

The credibility of public signals shapes this information environment. Financial disclosure lowers the cost of acquiring reliable information when investors regard the discloser as credible; weak or non-credible signals raise verification costs and the cost of capital (Diamond and Verrecchia, 1991; Verrecchia, 2001; Healy and Palepu, 2001; Leuz and Verrecchia, 2000; Bushman and Smith, 2001; Goldstein and Yang, 2017). Investors' updates also depend on the perceived bias and authenticity of the source (Dye, 1985; Cologna, Baumberger, Knutti, Oreskes, and Berthold, 2022; Koreh and Mandelkern, 2024). Misinformation can therefore affect liquidity without changing firms' underlying cash flows: it degrades the public signal, increases the cost of separating credible information from false claims, and changes the risk that liquidity providers attach to incoming orders. Recent studies linking digital information and misinformation to liquidity document related effects on market microstructure (Herzing and Mück, 2025; Dees and Ozyurt-Miller, 2026; Mo et al., 2026; Leippold, Wang, and Zhou, 2022).

2.2 Climate Misinformation and Financial Markets

Climate misinformation is a politically produced distortion of that public signal. Organized campaigns by carbon-intensive industries have manufactured uncertainty about climate science and regulation, while partisan media and digital platforms amplify the resulting claims (Oreskes and Conway, 2010; Farrell, 2015; Farrell, 2016; Supran, Rahmstorf, and Oreskes, 2023; Vosoughi, Roy, and Aral, 2018; Acemoglu, Ozdaglar, and ParandehGheibi, 2010). Exposure changes factual beliefs, hardens partisan misperceptions, erodes trust, and resists correction (Flynn et al., 2017; Nyhan and Reifler, 2010; Lewandowsky, Ecker, and Cook, 2017; Bursztyn et al., 2020; Lazer, Baum, Benkler, et al., 2018; Cobb, 2025). Climate misinformation is a political problem because it distorts the evidence citizens use to evaluate policy proposals and hold decision-makers accountable. Organized interests can exploit uncertainty about climate science to weaken support for mitigation and delay regulation (Farrell, McConnell, and Brulle, 2019). Understanding climate change requires interpreting long-term trends and probabilistic evidence; misleading arguments can exploit that difficulty by treating a cold weather event as a refutation of warming or uncertainty as evidence that action is unnecessary. Cook, Ellerton, and Kinkead (2018) show how such claims rely on faulty premises and reasoning. Confusion therefore need not reflect a simple absence of information: apparently plausible arguments and trusted partisan sources can make false claims persuasive. These studies establish the political and behavioral consequences of misinformation, but they do not show how the same content affects the liquidity of firms whose values depend on climate policy.

For businesses, these distortions can cloud expectations about regulation, demand for low-carbon products, and the returns to long-lived investments. That connection is plausible because climate-transition firms are unusually exposed to contested political information. Subsidies, carbon prices, disclosure requirements, and phase-out timetables are set through political institutions and international negotiations. Political and policy uncertainty carries a risk premium and raises volatility (Pastor and Veronesi, 2012; Pástor and Veronesi, 2013; Baker, Bloom, and Davis, 2016), while partisan information and market sentiment can move prices independently of fundamentals (Azzimonti and Fernandes, 2023; Lei and Luo, 2023; De Long, Shleifer, Summers, and Waldmann, 1990). Climate risk, news, and disclosure are likewise reflected in carbon premia, green-bond pricing, divestment, and institutional holdings (Engle et al., 2020; Bolton and Kacperczyk, 2021; Hong, Li, and Xu, 2019; Pankratz et al., 2023; Krueger, Sautner, Tang, and Zhong, 2024; Flammer, 2021; Friede, Busch, and Bassen, 2015; Drahos, 2025). Misinformation differs from ordinary policy uncertainty because it corrupts the signal investors use to evaluate these exposures.

The verification problem is especially salient in fragmented climate governance, where public rules, transnational networks, private standards, taxonomies, and ratings offer competing signals whose authority can be politically contested or vulnerable to capture (Keohane and Victor, 2011; Hale and Held, 2018; Bodansky, Brunnée, and Rajamani, 2016; Rajamani and Brunnée, 2017; Abbott and Snidal, 2000; Büthe and Mattli, 2011). Misinformation then makes genuine sustainability information harder to distinguish from greenwashing (Delmas and Burbano, 2011; Marquis, Toffel, and Zhou, 2016). The unresolved question is whether politically produced climate misinformation changes the trading costs through which markets process that information. The next section derives the competing liquidity predictions.

3 Theory: Climate Misinformation and Market Liquidity

The theoretical construct is information asymmetry: one party holds material information that another lacks (Akerlof, 1970; J. E. Stiglitz, 2000; J. Stiglitz and Kosenko, 2024a; J. Stiglitz and Kosenko, 2024b). The primary dependent variable is not information asymmetry itself, but the firm-day percent quoted bid–ask spread used to observe its liquidity consequences. In the sequential-trade model of Glosten and Milgrom (1985), a market maker posts a bid and an ask without knowing whether the next order comes from an informed or an uninformed trader. The spread compensates for expected losses to informed counterparties: the greater the perceived informed-trading risk, the wider the quote required to break even (Easley and O'Hara, 1992; Hasbrouck, 1991). A wider spread indicates lower quoted liquidity and is consistent with greater adverse-selection pressure, although it can also reflect inventory risk and other trading costs.

Climate misinformation changes the public signal that investors use to value climate-exposed firms. Subsidy regimes, carbon prices, and regulatory timetables are political outcomes, and misinformation makes claims about those outcomes harder to evaluate (Pastor and Veronesi, 2012; Pástor and Veronesi, 2013; Delmas and Burbano, 2011; Marquis et al., 2016). Investors differ in their ability and willingness to verify these signals. The resulting heterogeneity can affect trading through two competing microstructure channels.

Under the adverse-selection channel, sophisticated investors are better able to identify the credible climate-pricing signal, while less-informed investors are more likely to trade on false claims. This divergence increases the informational advantage of sophisticated traders. A liquidity provider who expects greater losses when orders originate from better-informed counterparties widens the bid–ask spread and quotes less depth. Per-dollar price impact should move in the same direction because trades reveal more private information. The observable prediction is therefore wider spreads and higher price impact, both indicating lower liquidity (Grossman and Stiglitz, 1980b; Chordia, Roll, and Subrahmanyam, 2000; Han, Hirshleifer, and Walden, 2022; Bordalo, Gennaioli, and Shleifer, 2020). Here, sophistication refers to investors' capacity to verify information, not to a firm's climate performance. Larger or wealthier firms may nevertheless have better disclosure, stronger compliance capacity, and a more informed investor base, jointly affecting misinformation exposure and liquidity. Firm fixed effects and the market-capitalization control address stable firm differences and observed size changes, but do not eliminate time-varying differences in disclosure quality or regulatory preparedness. Figure 1 summarizes the proposed adverse-selection pathway.

Under the liquidity-fragility channel, viral misinformation may divert attention from accurate information and attract inexperienced retail traders into climate-exposed stocks (Barber and Odean, 2008; Peng and Xiong, 2006; Sims, 2003). If the uninformed share rises, the market maker's perceived probability that the next order is informed falls, narrowing the quoted spread. Yet this new order flow is shallow and noisy: each additional dollar traded moves the price by more, increasing price impact and weakening price efficiency. Liquidity fragility is this joint state in which trading appears cheaper at the quoted spread while the market's capacity to absorb informed flow deteriorates.

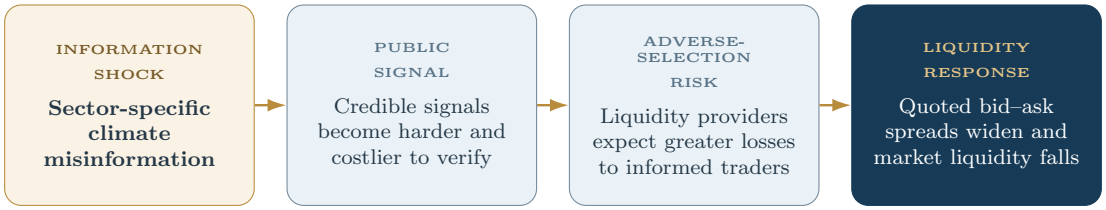


Figure 1: The adverse-selection pathway linking sector-specific climate misinformation to market liquidity.

The first two pathways imply different signs for the spread response. Adverse selection predicts wider quotes, whereas liquidity fragility predicts narrower quotes alongside weaker capacity to absorb informed trading. H1 therefore states that *higher security-day exposure to sector-specific climate misinformation is associated with narrower percent quoted spreads.* The statistical test uses the sign and precision of the spread coefficient across increasingly local comparisons. The positive coefficient in a baseline specification counts against H1 in that specification; an imprecise coefficient in a more saturated model does not justify a general rejection. The daily VWAP–midquote gap is examined as a complementary price-dispersion measure, but does not measure per-dollar transaction price impact and cannot complete a joint mechanism test.

Rolling CAPM beta and Fama–French five-factor excess returns provide benchmark outcomes rather than alternative measures of liquidity (De Long et al., 1990; Pastor and Veronesi, 2012). The climate-finance literature implies that financial responses depend on which climate news and disagreement is relevant to a security (Engle et al., 2020; Bolton and Kacperczyk, 2021; Hong et al., 2019; Pankratz et al., 2023). Their interpretation also depends on an appropriate regional factor mapping. The supplied factor-derived outcomes are retained as descriptive diagnostics, with their mapping limitation stated explicitly; they are not used to establish changes in systematic risk or return premia. Liquidity, timing and sector-condition comparisons supply the substantive empirical tests.

4 Data

The unit of analysis is the security-day. The financial input contains 726,890 observations on 204 listed climate-transition securities from January 2012 to December 2025. These securities—fossil-fuel incumbents and renewable challengers—are selected from a broader 19,183,738-security-day universe using nine TRBC energy sub-industries. The financial panel supplies daily quotes, trading characteristics and descriptive risk-and-return variables from Refinitiv Workspace, FRED, the World Bank and the Kenneth French Data Library. The climate-misinformation corpus supplies the 217,954 unique English-language Facebook posts used to construct the exposure measure. The observed index covers 3 January 2012–13

October 2025, and dates outside that support are excluded rather than coded as unexposed. Appendix Tables 18–22 and 23 describe the corpus and its classifications. Appendix Tables 6–11 document the supplied financial assembly, while the estimation-sample tables distinguish those inputs from the complete-case regressions. Processing is implemented in Python, following the transparency principles of Christensen and Miguel (2018).

4.1 Financial Panel and Outcome Measures

The financial data come from a panel of 204 listed climate-transition securities sourced from Refinitiv Workspace via the Eikon Python API. The sample is defined by membership in nine TRBC energy subindustries spanning the fossil-fuel value chain (Coal, Integrated Oil & Gas, Oil & Gas Drilling, Oil & Gas Exploration and Production, Oil & Gas Refining and Marketing, Oil & Gas Transportation Services, Oil Related Services and Equipment) and the renewable transition (Renewable Energy Equipment & Services, Renewable Fuels). Actual listing markets include mainland China, Hong Kong, Taiwan, Japan, Indonesia, Malaysia, Brazil, the United States, the United Kingdom, France, Germany and the Netherlands. Security identifiers are kept distinct: multiple share classes and cross-listings do not become independent firms. Daily prices, volume, turnover, capitalization, dividends, bid–ask quotes and VWAP describe the trading environment (Karpoff, 1987; Parkinson, 1980). The percent quoted spread is the bid–ask difference divided by the quote midpoint and expressed in percent. It is the primary liquidity outcome; it does not identify effective execution costs or the private-information component of the spread on its own.

One practical concern is timing. The Shanghai trading day ends before the New York session begins, so a same-date merge can associate posts made after an Asian close with that day's quotes. The timing checks distinguish the previous calendar day from the previous observed session with positive volume and a positive close. Session lags are formed before selecting complete cases on the spread, and source dates outside the corpus window or gaps exceeding ten calendar days are excluded. Static characteristics are merged by security. Listing market is assigned using the exchange suffix of the RIC, separating, for example, Indonesian .JK, Brazilian .SA and Taiwanese .TW listings from US and Japanese markets. The complete market crosswalk accompanies the replication materials. This assignment supplies the local market-day fixed effects, while headquarters country remains a separate characteristic.

The financial input also contains regional macroeconomic series from FRED and the World Bank: GDP growth, the supplied unemployment-labelled series, and CPI inflation (Federal Reserve Bank of St. Louis, 2025; World Bank, 2025). These are carried forward from lower-frequency observations (J. Stock and Watson, 2008):

$$M_{r,d} = M_{r,q(d)}, \qquad q(d) = \max\{q : \text{start}(q) \leq d\}. \tag{1}$$

The supplied region groups are not equivalent to actual listing markets. The substantive spread specifications therefore use market-day effects instead of relying on this regional macro mapping. In particular, the unemployment-labelled input is documented without assigning it a structural unemployment interpretation.

Daily Fama–French five-factor returns—market excess return (MktRF), size (SMB), value (HML), profitability (RMW), and investment (CMA)—are supplied from the Kenneth French Data Library.[1] The input includes regional factors and corresponding risk-free rates. Their descriptive coverage is reported in the appendix. Factor-derived outcomes require a correct association between each security and its relevant market. They are therefore retained as supplied-data benchmarks, with no claim that correcting a fixed-effect market label reconstructs a valid risk-free-rate series or regional factor exposure.

Three variables are constructed from the raw panel. The daily simple return is

$$R_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}} \times 100, \tag{2}$$

where $P_{i,t}$ is the closing price of firm i on day t. The daily excess return is $R^e_{i,t} = R_{i,t} - R_{f,t}$, where $R_{f,t}$ is the regional daily risk-free rate from the Fama–French data. The rolling CAPM beta comes from 252-trading-day rolling OLS regressions of firm excess returns on the regional market excess return, requiring a minimum of 126 observations per window:

$$\hat{\beta}_{i,t} = \frac{\text{Cov}(R^e_{i,s}, \text{MktRF}_s)}{\text{Var}(\text{MktRF}_s)}, \quad s \in [t-251, t]. \tag{3}$$

The rolling beta is a slow-moving measure by construction, and thin or non-synchronous trading can bias a conventional OLS beta (Scholes and Williams, 1977; Dimson, 1979). These issues matter alongside

[1] Data available at https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

market matching: a low estimated beta need not imply low fundamental climate risk. Appendix Table 12 reports supplied betas by stored region group and capitalization quintile as a description of the input. Since those groups do not identify actual listing markets, neither their differences nor the factor-derived regression coefficients establish cross-country differences in systematic risk. The quoted-spread analysis instead uses directly observed quotes and separately assigned market-day effects.

The merged climate-transition panel is uniquely indexed by security and date, with no duplicates. Mainland China and Indonesia contribute the largest security counts, followed by the United States. Appendix Table 13 documents the full financial input; Table 16 reports actual listing markets in the 485,800-observation prior-control sample, while the nested contemporaneous-control comparison has 486,655 observations. The prior-control sample contains 202 securities and 197 distinct company names. Missing quotes, nonpositive trading activity and required financial controls determine complete-case coverage. Retaining a long calendar panel does not itself eliminate survivorship or selection concerns.

Finally, the climate-transition panel is merged on calendar date and firm sub-industry with the firm-day misinformation-exposure index built from the Meta Content Library, described in Sections 4.2 and 4.3 below. The main regressions use $\text{Exp}_{i,t}^{\text{mean}}$ as the headline regressor of interest, with $\log(1 + \text{Exp}_{i,t}^{\text{sum}})$, $M_{i,t}^{\text{mean}}$, and $R_{i,t}^{\text{mean}}$ as variant specifications, all defined in Section 4.3.

4.2 Climate Misinformation Corpus

The misinformation measure is built from 217,954 unique English-language public Facebook posts—the subset of a 265,652-post corpus extracted from the Meta Content Library (MCL) whose Meta-recorded creation date matches at least one trading day in the climate-transition panel. The corpus was assembled through keyword searches across eight thematic datasets covering climate adaptation, climate politics, global climate justice, resilience, climate activism, and ecological themes; the posts span December 2009 to October 2025 and come from a long-tailed distribution of pages and profiles. Each post carries platform-reported engagement counts—likes, shares, comments, total reactions, and views—plus metadata on content type, surface type, and owner characteristics. Appendix Table 18 reports summary statistics on the corpus as a whole.

The corpus is an English-language, keyword-selected sample of public content, not a representative sample of national Facebook use. The available summaries do not establish the countries of origin of posts or the geographic distribution of their audiences, so differences in platform access, language, and user composition cannot be separated from differences in misinformation production. The exposure measure instead treats Facebook as a potential transnational information environment: claims about energy technologies, carbon policy, or fossil-fuel demand can circulate across borders and inform investors trading firms in the same sub-industry. For example, a claim originating in India could matter for a French-listed company if it reaches that company's investors or concerns markets or technologies relevant to its cash flows. The link is shared sector relevance and possible cross-border circulation, not the automatic transmission of every foreign post to every firm. The index measures the availability of relevant content, rather than observed investor attention or a country-to-country transmission effect.

The corpus enters this paper at the level of individual posts. Each (post, sub-industry) pair is scored by a single large language model on two continuous 0–100 scales—sector relevance and climate-or-energy misinformation—as described in Section 4.3; the firm-day index used in the empirical models is built from those scores. Three properties of the corpus matter for the analysis. First, the corpus is heavy-tailed in volume by year, with English-language climate-post volume more than tripling after 2019 (Appendix Table 19). Second, the corpus is concentrated in supply: a small set of high-volume pages and profiles produces a large share of total posts (Appendix Table 21), and platform-reported engagement counts are themselves heavy-tailed, with the highest-engagement posts diffusing on the order of three times the mean shares of the median post—a pattern consistent with the finding that misleading content travels disproportionately (Vosoughi et al., 2018). Third, the corpus mixes climate-scientific content, climate-political content, and climate-adjacent content that is on-topic for some sub-industries and entirely off-topic for others; the firm-day construction in Section 4.3 is what makes the sub-industry partition operational.

The classifier uses an open-weight model, and the post-level classifications and firm-day index built from them are released with the replication package. The raw post text falls under the MCL terms of use and is not redistributed. The classifier is assessed with a rubric-development spot-check (Section 4.3), which informed the rubric and is not an independent held-out or hand-coded gold-standard sample (Grimmer, Roberts, and Stewart, 2022). Constructing a gold standard is less straightforward than for a binary label because the task requires two continuous 0–100 judgments—sector relevance and climate-

or-energy misinformation—over nuanced content. A hand-coded benchmark would nevertheless provide a valuable direction for further measurement validation.

4.3 LLM Classification of Misinformation

This section describes the classification framework that turns the corpus into a firm-day measure of misinformation exposure (Wardle and Derakhshan, 2017). The design goal follows directly from the inference problem in Section 5: a regressor that is to vary across firms within a calendar day must be built by asking, for every post, a separate question of every sub-industry the panel contains. That is what the classifier does. It is called once on each of the 1,959,707 (post, sub-industry) pairs—covering the nine TRBC energy sub-industries and 217,954 posts with valid output—and returns, for each pair, how relevant the post is to that sub-industry and how much climate-or-energy misinformation it contains. A market-wide daily index that collapsed a model's per-post scores into one number would, by construction, assign every firm the same value on a given day and reinstate the firm-invariant problem this paper is designed to avoid; only the per-sub-industry judgment produces the cross-sectional variation the design needs.

A single high-capacity model is used rather than an ensemble. At 1.96 million judgments, the 35-billion-parameter model is tasked with distinguishing sincere false claims from satire and sector-relevant claims from off-topic content using one explicit rubric. Independent human annotation remains necessary to quantify classification accuracy. Automated detection systems are themselves vulnerable to adversarial manipulation and raise ethical concerns (Qiu, Liu, Zhou, and Huang, 2022; Horne, Nevo, and Smith, 2023), so the design relies on one locally hosted classifier with fixed decoding settings and persisted outputs and whose rubric is fixed in advance and made explicit below.

The model is `Qwen3.6-35B-A3B-FP8`, a mixture-of-experts model with 35 billion total parameters and approximately 3 billion active parameters per token, served in 8-bit floating-point precision using the vLLM 0.21 inference engine. Inference uses temperature zero and structured-JSON grammar enforcement via the `xgrammar` backend, with persisted structured outputs for each (post, sub-industry) judgment (Westwood, Grimmer, and Hall, 2025; Gilardi, Alizadeh, and Kubli, 2023; Santurkar, Durmus, Ladhak, et al., 2023). Each judgment receives the firm's sub-industry profile (industry name, the set of countries and listing regions of firms in that sub-industry, and a list of example firm names) and the post (text, date, content type, link-attachment caption and description). The rubric instructs the model to score two continuous 0–100 quantities: *sector relevance* $R_{p,s}$, capturing how materially the post's content relates to firms in sub-industry s, and *climate-or-energy misinformation* $M_{p,s}$, capturing the degree to which the post contains false, misleading, or pseudoscientific claims about climate science or energy.

Two features of the rubric matter. First, misinformation is bounded to climate or energy claims. A post that contains political conspiracy theories unrelated to climate—antisemitic content about EU politicians, anti-vaccination conspiracies, electoral fraud claims—scores $M_{p,s} = 0$ regardless of how false those other claims are; the score is a measure of climate-misinformation exposure, not a general misinformation index. Second, satire and parody are bounded to be not misinformation. A sarcastic Facebook post mocking climate denial that begins "MDPI, the predatory journal of which nothing can be believed, according to Facebook climate experts" receives $M_{p,s} = 0$ with the rationale "satirical parody mocking climate denialism, not a sincere assertion of false claims". These features were added to the rubric after a rubric-development spot-check on the first 2,000 judgments revealed that a default rubric over-scored political and parody content; the spot-check is reported in Appendix E.

The per-post composite exposure index is

$$\mathrm{Exp}_{p,s} \;=\; \frac{R_{p,s}\, M_{p,s}}{100}, \tag{4}$$

on a 0–100 scale, which is non-zero only when the post is sector-relevant for sub-industry s *and* contains climate or energy misinformation. The product, rather than either factor alone, is the object of interest because the economic channel requires both: a false climate claim that has nothing to do with coal does not change what a coal trader needs to price, and an accurate, highly relevant post about coal carries no misinformation for the market to misprice. The index isolates the intersection—misinformation that is *about* the firm's sector. The firm-day regressor used in the empirical models is the mean of this composite across the day's paired posts, evaluated at the firm's sub-industry:

$$\mathrm{Exp}^{\mathrm{mean}}_{i,t} \;=\; \frac{1}{|P_t|} \sum_{p \in P_t} \mathrm{Exp}_{p,\, s(i)}, \tag{5}$$

where P_t contains the valid scored climate posts for the relevant sector whose Meta-recorded creation date matches calendar date t, and $s(i)$ is the TRBC sub-industry of firm i. Two volume-style variants enter the robustness checks: $\text{Exp}_{i,t}^{\text{sum}} = \sum_p \text{Exp}_{p,s(i)}$ (sensitive to the number of climate posts a firm's sector sees on a given day) and $\log(1 + \text{Exp}_{i,t}^{\text{sum}})$ (which tames the right skew). The raw components $R_{i,t}^{\text{mean}}$ and $M_{i,t}^{\text{mean}}$ are also persisted, to test whether the composite or one of its constituents carries the signal.

Three properties of the regressor matter for the inference design. First, two firms in different sub-industries on the same date receive different exposures: the model evaluates each post against the specific sub-industry, so a Renewable-Fuels firm and a Coal firm on the same trading day inherit different scores from the same set of posts. Second, two firms in the *same* sub-industry on the same date share a value by construction; the level of aggregation at which the model is asked to reason is the sub-industry, not the firm. Firm-level fixed effects in the empirical models absorb the constant level of exposure that any firm experiences over the sample, leaving within-firm between-day variation as the identifying contrast. Third, the share of firm-days with exactly $\text{Exp}^{\text{mean}} = 0$ is 44% in the climate-transition panel (Appendix Table 24 and Appendix Figure 7)—these are firm-days on which paired climate posts were either off-topic for the firm's sub-industry or contained no misinformation—so the regressor is sparse, and the volume variants in the robustness section capture how the number of paired posts mediates the response. A total of 398 of the 1,959,707 (post, sub-industry) judgments returned a parsing error after five retries (0.02%) and are dropped from any downstream firm-day on which they would otherwise have entered the average.

5 Empirical Strategy

The empirical strategy separates the main quoted-liquidity design from benchmark risk-and-return diagnostics. The baseline tests the spread prediction using within-security changes in the sector-specific misinformation index. A nested comparison then introduces daily and local market-day effects, followed by sector-month effects. These comparisons make the identifying variation explicit rather than allowing a change in sample to explain a coefficient difference. The factor-derived diagnostics remain secondary because the supplied regional mapping limits their financial interpretation.

5.1 Design 1: Market Liquidity

The baseline spread specification is:

$$\text{SpreadPercent}_{i,t} = \beta_1 \text{Exp}_{i,t}^{\text{mean}} + \boldsymbol{\gamma}' \mathbf{X}_{i,t} + \alpha_i + \delta_{q(t)} + \varepsilon_{i,t}. \tag{6}$$

Here $\mathbf{X}_{i,t}$ contains log one plus trading volume, log market capitalization and the daily high–low price range divided by the closing price and expressed in percent. α_i denotes security fixed effects and $\delta_{q(t)}$ year-month effects. All logged variables are natural logarithms. Continuous financial controls and outcomes are winsorized at the first and ninety-ninth percentiles; exposure scores are not winsorized. The quoted spread is reconstructed as $200(A - B)/(A + B)$, where A and B are the daily ask and bid. Positive bid, non-crossed positive spread, positive volume and positive close are required.

The nested comparison replaces year-month effects first with calendar-date effects and then with listing-market-by-date effects. The latter absorb daily news, trading conditions and other shocks shared by securities in the same local market. All four contemporaneous-control specifications use the same 486,655 observations, controls and tail caps. The final nested specification adds sub-industry-by-month effects, separating daily exposure variation from the monthly information and valuation environment of the sector:

$$\text{SpreadPercent}_{i,t} = \beta_1 \text{Exp}_{i,t}^{\text{mean}} + \boldsymbol{\gamma}' \mathbf{X}_{i,t} + \alpha_i + \lambda_{m(i),t} + \mu_{s(i),q(t)} + \varepsilon_{i,t}. \tag{7}$$

The fully conditional model additionally replaces contemporaneous financial controls with their values at the previous observed active trading session, denoted $\mathbf{X}_{i,t^-}$. This comparison uses 485,800 observations. It reduces direct conditioning on current trading activity, which may itself respond to the information environment, but does not make exposure exogenous. Its different control timing is shown separately from the common-sample nested comparison.

The companion price-dispersion outcome is $100|\text{VWAP}_{i,t} - \text{MidPrice}_{i,t}|/\text{MidPrice}_{i,t}$. The supplied fields called OrderFlow and PriceImpact equal the signed VWAP–midpoint difference and its absolute percentage transformation. These identities do not recover signed traded quantities or per-dollar transaction price impact. The gap regression therefore describes another aspect of prices without identifying the information content of a trade. Variant regressors separate composite exposure, its volume transformation, average misinformation, average relevance and positive-content frequency.

5.2 Design 2: Benchmark Risk and Return Outcomes

5.2.1 Model II: Rolling CAPM Beta

The systematic-risk benchmark relates the supplied rolling CAPM beta to exposure:

$$\hat{\beta}_{i,t} = \lambda \mathrm{Exp}_{i,t}^{\mathrm{mean}} + \boldsymbol{\gamma}'\mathbf{X}_{i,t} + \alpha_i + \delta_{q(t)} + \varepsilon_{i,t}. \tag{8}$$

The beta is computed over 252 trading observations with at least 126 observations per window. In principle, λ asks whether information conditions co-move with the security's covariance with a relevant market factor. In the supplied input, the regional factor assignment is not equivalent to the corrected listing-market crosswalk. Accordingly, the saved estimates are descriptive diagnostics and cannot identify a climate-misinformation response in correctly matched systematic risk. Their null findings are retained without treating them as proof of unchanged fundamental risk.

5.2.2 Model III: Augmented Fama–French Five-Factor Model

The Fama–French benchmark decomposes excess returns as:

$$R_{i,t}^{e} = \beta_{m,i}\mathrm{MktRF}_t + \beta_{s,i}\mathrm{SMB}_t + \beta_{v,i}\mathrm{HML}_t + \beta_{p,i}\mathrm{RMW}_t + \beta_{c,i}\mathrm{CMA}_t + \varepsilon_{i,t}. \tag{9}$$

The augmented specification introduces the exposure index:

$$R_{i,t}^{e} = \sum_k \beta_k F_{k,t} + \phi \mathrm{Exp}_{i,t}^{\mathrm{mean}} + \boldsymbol{\gamma}'\mathbf{X}_{i,t} + \alpha_i + \delta_{q(t)} + \varepsilon_{i,t}. \tag{10}$$

The factors capture market excess return, size, value, profitability and investment (Fama and French, 1993; Fama and French, 2015). A coefficient on exposure in this regression is a conditional return association, not by itself a test that misinformation is a priced risk factor. The supplied factor mapping imposes the same diagnostic limitation as Model II. These benchmarks retain their role in distinguishing the research question about trading conditions from a claim about expected-return premia.

5.3 Identification Strategy

Identification in the baseline rests on within-security, within-month variation. Security effects absorb time-invariant characteristics, including sub-industry, listing market and stable disclosure differences. Month effects absorb shocks common to that calendar month. The sector-specific exposure construction also supplies cross-sectional variation within a date. That variation permits market-day fixed effects without absorbing the index completely. Sector-day effects would absorb it, since securities in the same sector and date inherit the same value. The index therefore makes a local comparison feasible while also defining the limit of what the panel identifies.

Sector-month effects change the economic comparison rather than simply adding another precision adjustment. Without them, the coefficient can reflect a sector's sustained information and valuation conditions within the month. With them, it is identified from more transitory deviations in that environment. A change in the coefficient can arise because those controls remove confounding sector conditions or absorb a persistent information channel. The observational data do not distinguish these explanations. The positive broader association and the null saturated association must therefore be reported together.

Inference reflects assignment at the sub-industry/day level. Standard errors cluster by sub-industry and calendar date, allowing dependence among different securities in a sector over time and among sectors on a date (Moulton, 1990; Petersen, 2009). With nine sector clusters, t_8 reference probabilities and confidence intervals are approximate (Cameron and Miller, 2015; MacKinnon, Nielsen, and Webb, 2023). Security/date clustering is reported alongside it. Date-score HAC with bandwidths of 5, 20 and 60 observed dates and an enumerated sector score-sign sensitivity provide additional reference calculations; the latter requires independent symmetric sector scores and is not exact randomization inference (Driscoll and Kraay, 1998). Full definitions appear in the supplement.

The main limitation is that no exogenous shock separates exposure from daily sector-specific news, policy developments or changing investor beliefs. Lagging content improves the description of temporal ordering but does not remove persistent confounding. A future-content lead is informative about timing specificity without constituting a causal test. Relevance subsets similarly assess measurement composition, rather than supplying an externally validated placebo. The analysis uses these distinctions to establish where the positive spread association is supported and where it remains unresolved.

6 Results

The results connect sector-specific climate misinformation to quoted trading conditions in the climate-transition sample. The central positive finding is that the association with wider spreads survives local market-day controls. The next comparison establishes its boundary: adding sector-month effects changes the sign and leaves an imprecise coefficient. Both findings are reported in Table 1, using the completed estimates and pyfixest 0.40.1. The nested contemporaneous-control models share a sample, so the coefficient change cannot be attributed to changing observations or tail caps.

Table 1: Exposure and quoted spreads across fixed-effect specifications

Specification	Coefficient	SE	p	N
Security + month	+0.0041	0.0020	0.073	486,655
Security + date	+0.0145	0.0079	0.102	486,655
Security + market-day	+0.0129	0.0050	0.033	486,655
+ sector-month	-0.0033	0.0019	0.117	486,655
Sector-month + prior controls	-0.0014	0.0022	0.533	485,800

The first four rows use identical observations, contemporaneous controls and tail caps. The final row uses prior-session controls. The outcome is the quoted spread in percentage points; uncertainty clusters by sector/date, with nine sectors. Market-day effects use actual listing markets. The positive and imprecise estimates are retained together.

6.1 Model I: Misinformation and Quoted Spreads (H1)

Greater sector-relevant climate misinformation is positively associated with quoted spreads under security/month effects (+0.0041, sector/date $p = 0.073$). The sign remains positive with date effects (+0.0145, $p = 0.102$) and with correctly mapped local market-day effects (+0.0129, $p = 0.033$). The latter result is especially informative for the question posed by the paper: securities with different sector content scores face different quoted trading conditions even after their shared local trading environment is absorbed. It is a conditional association rather than an identified effect, but it shows that the positive pattern is not confined to a specification that leaves all daily market-wide movements uncontrolled.

Table 2: Model I: Quoted spreads with local market-day controls

Variable	Coefficient	SE	p
Mean sector-specific exposure	+0.0129	0.0050	0.033
Log one plus volume	-0.2952	0.0315	< 0.001
Log market capitalization	-0.0941	0.0384	0.040
Daily price range (percent)	+0.0676	0.0086	< 0.001

$N = 486{,}655$. Security and listing-market/date effects are absorbed. Financial variables are winsorized at the first and ninety-ninth percentiles; content scores are unchanged. SEs cluster by sector and date, with approximate t_8 reference probabilities. This model does not absorb sector-month conditions; that comparison is shown explicitly in Table 1.

The financial control coefficients are consistent with familiar liquidity relationships. In the market-day model, log trading volume is negatively associated with spreads (-0.2952, $p < 0.001$), capitalization also enters negatively (-0.0941, $p = 0.040$), and the daily price range enters positively (+0.0676, $p < 0.001$). More active trading and larger capitalization are associated with tighter quotes, while a larger range is associated with higher trading costs (Stoll, 2000; Chordia et al., 2000; Amihud and Mendelson, 1986). These are partial associations and do not isolate inventory costs or the private-information content of order flow (Kyle, 1985; Easley and O'Hara, 1996; Hasbrouck, 1991). The result supports using the spread as a direct description of quoted liquidity while keeping its components conceptually distinct.

Sector-month controls materially qualify the exposure result. On the same 486,655 observations, the content coefficient becomes -0.0033 ($p = 0.117$). The fully conditional model with prior-session controls yields -0.00143, sector/date standard error 0.00220, $p = 0.533$, and an approximate 95% interval of $[-0.00650, 0.00364]$. It uses 485,800 observations on 202 securities from 197 distinct company names.

These estimates do not establish an independent daily spread-widening response within the sector's monthly environment. Equally, they do not establish that the broader positive association is spurious: the sector-month effect can absorb sustained information conditions as well as confounding valuation changes. The empirical finding is a distinction between those two comparisons, not a choice to suppress one of them.

The economic scale is modest. The fully conditional sample's content standard deviation is 0.710, and its winsorized mean spread is 0.473%, or 47.29 basis points. Scaling the fully conditional coefficient by that content standard deviation gives -0.102 basis points, with an approximate interval of $[-0.462, 0.258]$. This scaling describes a fitted association; it is not the consequence of exposing an investor to misinformation. The index's standard deviation after residualizing the controls and fixed effects is 0.297, showing that the local comparison retains measurable content variation even though it yields an imprecise coefficient.

For H1, the positive monthly and market-day coefficients run against the narrower-spread prediction and are compatible with the adverse-selection account in Section 3 (Glosten and Milgrom, 1985; Easley and O'Hara, 1992). The saturated estimates do not justify a general rejection of H1 or a definitive selection of adverse selection over participation. This distinction preserves the substantive positive result without claiming that a quoted-spread regression alone identifies why liquidity suppliers change their quotes.

6.1.1 Complementary Price Dispersion

The companion outcome is the absolute VWAP–midquote gap in percent, rather than per-dollar transaction price impact. Its exposure coefficient under the fully conditional specification is $+0.00105$ ($p = 0.760$; $N = 485{,}798$). The estimate supplies no separate statistically discernible gap response. Its positive point estimate is compatible with several information and trading-cost mechanisms but cannot identify adverse selection. No inference about market depth or the response of prices to a dollar of signed trading is drawn from this variable.

This measurement distinction matters to interpreting the theoretical argument. Quoted spreads describe the prices posted by liquidity suppliers; the daily VWAP gap describes the relation between a trade-price average and a quote midpoint. Neither its sampling time nor an individual order's direction is identified here. The two measures can be reported together as descriptions of the trading environment, but do not constitute the spread/transaction-impact test required to separate the mechanisms uniquely. The null gap finding is therefore retained as a limit on mechanism evidence, not recast as support for either channel.

6.1.2 Lag Specification

Table 3 compares contemporaneous exposure, the previous observed active session, the exact previous calendar day, and the next active session on a common sample of 388,227 observations. The session-grid construction retains the distinction between a Friday-to-Monday transition and an exact calendar-day join. No session lag is formed across a source date outside the content window or a gap exceeding ten calendar days.

Table 3: Timing comparison on a common sample

Content timing	Coefficient	SE	p
Current content	0.0001	0.0024	0.962
Previous active session	-0.0016	0.0021	0.468
Previous calendar day	-0.0017	0.0017	0.330
Next active session	0.0078	0.0047	0.135

Each model uses 388,227 observations and the principal fixed effects and prior controls. Exposure outside the corpus window is unavailable; session gaps over ten days are excluded. Sector/date inference.

In the common sample, current exposure has coefficient $+0.0001$ ($p = 0.962$), previous-session exposure -0.0016 ($p = 0.468$), and previous-calendar-day exposure -0.0017 ($p = 0.330$). The next-session lead is positive but imprecise ($+0.0078$, $p = 0.135$). On the larger available session-lag sample of 485,759 observations the coefficient is -0.0010 ($p = 0.652$). These checks do not establish an additional delayed widening response after local market-day and sector-month conditions are absorbed. They retain temporal

information without equating a lag with causal ordering. Serially correlated content and sector news can influence both leads and lags.

6.1.3 Sector Relevance and Content Composition

The climate-specific question requires distinguishing sector relevance from the misinformation component of content. The completed sensitivity estimates report the composite, log content sum, misinformation score, relevance score, positive-content indicator and the share scoring at least 25 on misinformation. Under the fully conditional specification, none establishes a positive spread association at conventional levels. Adding log post count and mean relevance jointly leaves the composite coefficient small and imprecise (-0.0013, $p = 0.591$). Table 27 in the appendix reports these findings without selecting a preferred content measure by its probability value.

A relevance-threshold subset changes both the posts entering an average and its denominator. For a high-relevance share q and conditional mean $\bar{e}^H$, its contribution to the all-post mean is $q\bar{e}^H$. A conditional-mean regression can therefore have a different sign from an all-post regression without an algebraic contradiction. If no post qualifies, the conditional mean is undefined even though its contribution is zero. Appendix E explains this distinction. The distributed inputs do not contain the post-ID/date crosswalk needed to reconstruct those thresholded daily series, so no unregenerated irrelevant-content placebo is used to claim specificity.

6.1.4 Subperiod Stability

The corpus becomes substantially larger after 2019, while measured exposure diverges across fossil-fuel and renewable sub-industries. Figure 2 documents that distributional pattern. It is substantively relevant to the information conditions facing energy securities: contested claims do not apply uniformly across technologies, operations or policy exposures. It motivates keeping the sector dimension in the content measure and in the econometric comparison.

The formal post-2019 slope interaction in the fully conditional model is -0.0019 ($p = 0.687$). The data therefore do not establish that the spread association strengthened after 2019 once market-day and sector-month conditions are accounted for. A change in corpus scale can affect composition and measurement precision without implying a changed financial response. The positive result under broader controls and the distributional divergence across sectors are accordingly reported as different findings, not treated as interchangeable evidence for a stronger post-2019 mechanism.

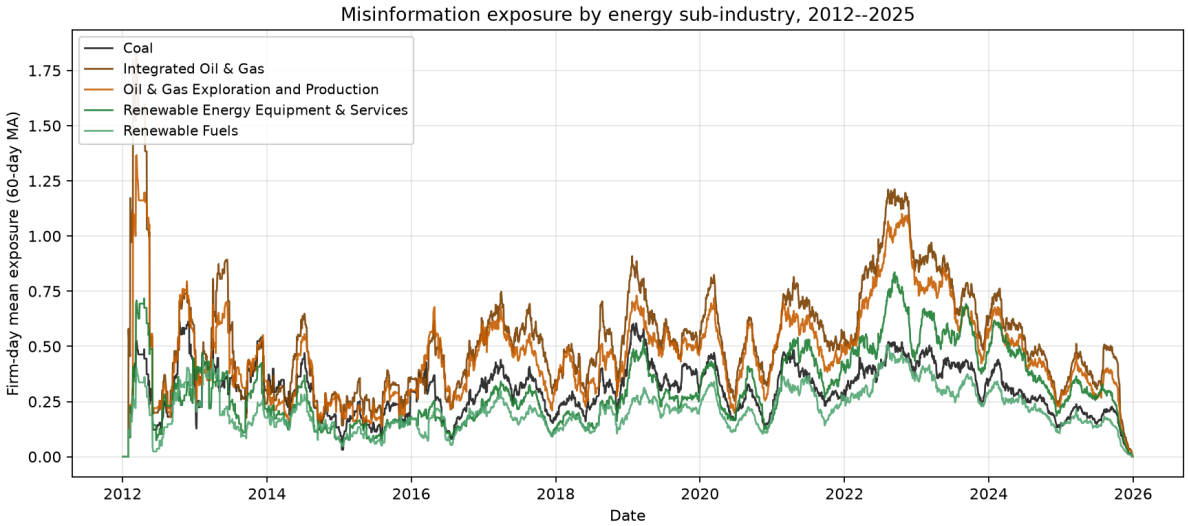


Figure 2: Mean measured exposure by climate-transition sub-industry (60-day moving average). Fossil-fuel and renewable sub-industries exhibit different content patterns. This distributional contrast does not itself establish a difference in spread-response coefficients.

6.2 Model II: Misinformation and Systematic Risk (CAPM Beta)

The saved factor-derived diagnostic has a small, imprecise coefficient on mean exposure (−0.0007, security/date $p = 0.53$, $N = 384{,}017$). The log content-sum and misinformation-score variants are also imprecise. Table 4 retains the full diagnostic readout so that the risk benchmark is transparent. The input's regional factor mapping is not equivalent to actual listing markets, however, and these coefficients are not presented as validated cross-market estimates of systematic risk. A null diagnostic cannot establish that misinformation leaves correctly measured covariance risk unchanged.

Table 4: Climate-Transition Panel, PanelOLS Full Readout: Model II (CAPM Beta)

	Exp^{mean}			$\log(1+\text{Exp}^{\text{sum}})$		
Term	Coef.	SE	p	Coef.	SE	p
Exposure term	−0.0007	0.0012	0.525	−0.0002	0.0005	0.716
Log Volume	+0.0336***	0.0056	<0.001	+0.0336***	0.0056	<0.001
Log Market Cap	−0.0398**	0.0200	0.048	−0.0398**	0.0200	0.048
Spread Percent	−0.0293**	0.0126	0.021	−0.0293**	0.0126	0.021
Dividend Yield	−0.0074**	0.0036	0.045	−0.0074**	0.0036	0.045
Intraday Volatility	+0.0037*	0.0020	0.072	+0.0037*	0.0020	0.072
CPI Inflation	−0.0217***	0.0065	0.001	−0.0217***	0.0065	0.001
GDP Growth	−0.0425***	0.0097	<0.001	−0.0425***	0.0097	<0.001
Unemployment	−0.0297***	0.0042	<0.001	−0.0297***	0.0042	<0.001
Observations			384,017			384,017
R^2			0.671			0.671
Within R^2			0.052			0.052
Adj. Within R^2			0.052			0.052
Wald F (joint)			14.26			14.26
Firm FE			Yes			Yes
Year-Month FE			Yes			Yes
SE clustering			Firm + Date			Firm + Date

Scope: Supplied factor-derived diagnostic. The regional factor mapping is not validated by the listing-market correction, so this table does not establish cross-market risk or return effects.

Notes: Dependent variable: 252-day rolling CAPM beta, Equation 8, 204-firm climate-transition panel. The two column blocks replace the exposure term with the firm-day mean exposure $\text{Exp}^{\text{mean}}_{i,t}$ and its concave volume variant $\log(1+\text{Exp}^{\text{sum}}_{i,t})$; control coefficients are nearly identical across the two blocks, as expected by Frisch–Waugh–Lovell when two variants of a sparse firm-day regressor have negligible partial correlation with the controls. Standard errors are CRV1 two-way clustered by firm (RIC) and calendar date. Continuous controls are winsorized at the 1st and 99th percentiles; the exposure regressors are not winsorized. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The distinction is consistent with the different time scales involved. A rolling 252-observation beta changes slowly, whereas the content index and quoted spreads vary daily. Volume, size, dividends, volatility and economic conditions can affect a rolling estimate in ways that differ from their association with a daily quote (J. H. Stock and Watson, 2002; Baker et al., 2016). The benchmark therefore helps define the scope of the paper: the positive evidence concerns quoted trading conditions under specified controls, rather than a general finding about changes in market risk.

6.3 Model III: Fama–French Excess Returns

The supplied excess-return diagnostic is negative for mean exposure (−0.0224) but loses precision when security and date are both clustered ($p = 0.151$, $N = 382{,}890$). Firm-only clustering produces a smaller standard error without changing the coefficient. Table 5 retains this comparison. The probability-value difference illustrates why date-level dependence matters; it does not establish a return effect. The supplied regional factor and risk-free-rate mapping further limits a cross-market excess-return interpretation.

The benchmark provides no independent return-pricing claim. More generally, a daily content coefficient added to a factor regression need not identify a priced risk factor, and an insignificant coefficient need not imply that information quality has no valuation consequences. Keeping risk, returns and quoted

Table 5: Climate-Transition Panel, PanelOLS Full Readout: Model III (Fama–French Excess Return)

	Exp^{mean}			$\log(1+\text{Exp}^{\text{sum}})$		
Term	Coef.	SE	p	Coef.	SE	p
Exposure term	-0.0224	0.0156	0.151	-0.0048	0.0089	0.591
FF_MktRF	$+0.6668^{***}$	0.0439	<0.001	$+0.6667^{***}$	0.0439	<0.001
FF_SMB	$+0.2003^{***}$	0.0450	<0.001	$+0.2002^{***}$	0.0450	<0.001
FF_HML	$+0.4195^{***}$	0.0481	<0.001	$+0.4196^{***}$	0.0481	<0.001
FF_RMW	$+0.0066$	0.0345	0.848	$+0.0072$	0.0346	0.835
FF_CMA	$+0.0169$	0.0446	0.704	$+0.0171$	0.0446	0.702
Log Volume	$+0.1516^{***}$	0.0159	<0.001	$+0.1517^{***}$	0.0159	<0.001
Log Market Cap	$+0.0677^{***}$	0.0215	0.002	$+0.0677^{***}$	0.0215	0.002
Spread Percent	$+0.1435^{***}$	0.0279	<0.001	$+0.1434^{***}$	0.0279	<0.001
Dividend Yield	-0.0143^{***}	0.0029	<0.001	-0.0142^{***}	0.0029	<0.001
Intraday Volatility	$+0.0252^{**}$	0.0108	0.021	$+0.0252^{**}$	0.0108	0.021
CPI Inflation	$+0.0008$	0.0119	0.945	$+0.0008$	0.0119	0.946
GDP Growth	$+0.0077$	0.0173	0.656	$+0.0077$	0.0173	0.658
Unemployment	-0.0068	0.0081	0.403	-0.0068	0.0081	0.400
Observations			382,890			382,890
R^2			0.077			0.077
Within R^2			0.069			0.069
Adj. Within R^2			0.069			0.069
Wald F (joint)			36.49			36.48
Firm FE			Yes			Yes
Year-Month FE			Yes			Yes
SE clustering			Firm + Date			Firm + Date

Scope: Supplied factor-derived diagnostic. The regional factor mapping is not validated by the listing-market correction, so this table does not establish cross-market risk or return effects.

Notes: Dependent variable: daily Fama–French five-factor excess return, Equation 10, 204-firm climate-transition panel. The two column blocks replace the exposure term with the firm-day mean exposure $\text{Exp}_{i,t}^{\text{mean}}$ and its concave volume variant $\log(1+\text{Exp}_{i,t}^{\text{sum}})$; control coefficients are nearly identical across the two blocks, as expected by Frisch–Waugh–Lovell when two variants of a sparse firm-day regressor have negligible partial correlation with the controls. Standard errors are CRV1 two-way clustered by firm (RIC) and calendar date. Continuous controls are winsorized at the 1st and 99th percentiles; the exposure regressors are not winsorized. $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$.

liquidity conceptually separate allows the positive spread association to be stated directly, while leaving the corresponding null and imprecise results visible.

7 Discussion

The findings support a substantive connection between the climate-information environment and quoted liquidity in energy equities. In the local market-day specification, a higher sector-relevant misinformation score is associated with wider quoted spreads. This is the direction predicted when traders differ in their ability to verify climate claims and liquidity suppliers face greater information risk. The positive estimate remains after shared local trading conditions are absorbed, giving the result more substance than an uncontrolled correlation between social-media activity and financial markets. Its interpretation is an association within a specified comparison, rather than a causal cost estimate.

The theoretical contribution is to bring politically produced misinformation into a framework usually applied to the distribution of financial information. Claims about climate science, carbon restrictions, subsidies or the viability of low-carbon technologies can be costly to verify. Investors with more expertise or better access to credible information may interpret those claims differently, giving liquidity providers a reason to adjust quotes (Glosten and Milgrom, 1985; Grossman and Stiglitz, 1980a; Goldstein and Yang, 2017). The data are consistent with that account under monthly and market-day controls. They do not observe a trader's verification decision, investor sophistication or order direction, so they do not identify that mechanism uniquely. Maintaining the distinction between a theoretical channel and its directly observed implications strengthens the contribution.

Sector-month controls locate the limit of the evidence. After those conditions are absorbed, the coefficient changes sign and is imprecise, both with contemporaneous and prior-session financial controls. This does not support a stable independent daily widening response. It also changes the economic object being estimated: shared monthly sector conditions may include genuine information-verification pressures alongside commodity-price changes, policy news and valuation uncertainty. Removing that component is a demanding local comparison, not a demonstration that the remaining coefficient alone measures the total information costs of climate misinformation. The paper therefore reports the positive broader association and the null local association together. Neither can substitute for a design that separates exogenous misinformation from the news environment in which it circulates.

The political-economy implication is most direct at the sub-industry level. Fossil-fuel chains and renewable technologies face different claims about demand, regulation and technological feasibility. A coal-security trader and a renewable-fuels trader can therefore face different relevant narratives on the same date. The sector-sensitive measure represents that distinction rather than assigning every security the same daily content value. The observed fossil–renewable exposure divergence is informative about the distribution of contested content, even though the formal post-2019 interaction does not establish a stronger liquidity response. This is a measurement contribution with substantive content, rather than merely a choice of regression fixed effects.

The wider literature gives that distinction practical significance. Misinformation can affect beliefs, trust and collective action, while energy investment depends on expectations about policy and technology (Cobb, 2025; Farrell, 2016; Engle et al., 2020; Bolton and Kacperczyk, 2021). A liquid secondary market can matter to financial conditions surrounding the transition, but this paper does not observe the cost of financing a new project or an issuer's investment decision. The evidence speaks to posted trading terms in listed equities. Connecting those terms to primary financing or real investment would require additional data and an identified transmission mechanism. The market-liquidity contribution is useful on its own without making that additional claim.

The corpus also highlights why the supply of misinformation and its audience should be kept distinct. The reach-based arguments of Acemoglu et al. (2010) and Vosoughi et al. (2018) motivate attention to diffusion, but the daily index averages classified content without weighting it by observed investor exposure. The supplied engagement and owner-network summaries document a heavy-tailed public information environment; they do not show which investors saw a post. Likewise, volume and relevance variants do not establish a positive coefficient in the fully conditional model. The appropriate interpretation is that sector-sensitive content captures a plausible information environment whose financial association depends on the comparison, not that a high post score constitutes a measured dose of investor misinformation.

Three limitations remain particularly relevant. First, the English-language, keyword-selected corpus has uneven geographic and temporal coverage. This matters in markets such as mainland China and Indonesia, which account for many sample securities, and cannot be solved by relabelling the content as globally representative. Second, rubric-development checks helped define the classifier but do not supply

independent precision, recall or human agreement. The content measure is reproducible from persisted scores, while its external accuracy remains a separate empirical question (Grimmer et al., 2022; Gilardi et al., 2023). Third, nine sector-level information series limit cluster-based inference despite the large number of security-day rows. The alternative covariance and leave-out checks describe that uncertainty transparently instead of treating sample size as independent information.

For research on the political economy of climate finance, these results provide a clear basis for further work. Sector-specific classification retains the political and economic content needed for a meaningful market comparison. Correctly assigned market-day effects establish that the positive association is not simply a common local-market movement. Sector-month controls then show that the evidence does not isolate an additional daily misinformation response. Those are complementary findings. An event study or instrument grounded in a defensible information shock, together with local-close timestamps and direct investor-flow data, would be needed to distinguish persistent information conditions from a causal trading-cost effect.

8 Conclusions

Climate misinformation is associated with wider quoted spreads in the monthly and local market-day specifications for climate-transition equities. The positive local market-day estimate establishes a connection between sector-relevant climate content and trading conditions after shared local daily shocks are absorbed. The finding is consistent with information-verification costs and supports bringing the political production of climate misinformation into research on market liquidity. Its empirical boundary is equally clear: adding sector-month controls changes the sign and leaves an imprecise coefficient, including in the model with prior-session financial controls.

Four conclusions follow. First, the construction of the content measure matters. Classifying posts separately against nine energy sub-industries preserves differences in the political information relevant to fossil-fuel and renewable securities. The resulting within-date variation makes a local market-day comparison possible. It does not create independent treatment at the level of each security, and inference must respect the sector assignment.

Second, the positive spread association is conditional on the sector comparison. It is statistically discernible under local market-day effects but does not persist as a robust positive coefficient after sector-month conditions are removed. The evidence therefore supports an association in the broader sector information environment while leaving a distinct daily causal response unresolved. The null result narrows the claim without requiring that climate misinformation be considered economically unimportant.

Third, the complementary VWAP-gap and timing checks provide no separate robust response under the fully conditional design. The gap is not per-dollar transaction price impact, and a lag is not an exogenous information shock. The risk and excess-return diagnostics are also inconclusive and constrained by their supplied factor mapping. They are retained as boundaries on the financial interpretation rather than used to amplify the main result.

Fourth, the corpus and sub-industry profiles document an uneven climate-information environment. The increase in content volume after 2019 and the fossil–renewable exposure divergence motivate sector-sensitive measurement, but the formal slope interaction does not establish a strengthened spread relationship. This distinction keeps a substantive distributional finding separate from a claim about changing financial sensitivity.

The paper contributes a sector-sensitive bridge between the political production of climate misinformation and the quoted trading environment of energy equities. Future work can build on that bridge through independent multilingual validation, exchange-local information timing, defensible exogenous events and direct measures of investor participation. Applying the same construction to other domains of politically contested knowledge—public health, technology and financial regulation—would also test how broadly the relationship extends (Lins, Servaes, and Tamayo, 2017; Taber and Lodge, 2006). The present evidence establishes where the positive liquidity association is observed and states clearly where the data do not support a stronger claim.

Ethics and data availability

The source research received institutional ethics approval for the Meta Content Library analysis. The replication materials supply the financial analysis, fixed classifications, saved estimates, source tables and figure assets. Raw Meta Content Library records and post text are access-controlled; financial

observations originate from licensed Refinitiv data. The financial robustness tables reproduce from the documented fixed inputs. Reconstructing the supplied corpus engagement summaries and owner-network figures requires authorized platform records. An author-identified repository link is supplied with the title page.

Declaration of generative AI and AI-assisted technologies

OpenAI Codex (GPT-6) assisted with analysis-code development, reproducible visualization and targeted manuscript editing. Numerical estimates are generated by the documented statistical computations from fixed inputs. Language-model classification is described as a research method in Section 4.3. Scientific interpretation, verification and submission remain the author's responsibility.

References

Abbott, K. W., & Snidal, D. (2000). Hard and soft law in international governance. *International Organization*, *54*(3), 421–456.

Acemoglu, D., Ozdaglar, A., & ParandehGheibi, A. (2010). Spread of (mis)information in social networks. *Games and Economic Behavior*, *70*(2), 194–227.

Akerlof, G. A. (1970). The market for lemons: Quality uncertainty and the market mechanism. *Quarterly Journal of Economics*, *84*(3), 488–500.

Aliyev, K., Aquilina, M., Razayev, K., & Zhu, S. (2025). Through stormy seas: How fragile is liquidity across asset classes and time? *BIS Working Paper 1229*.

Amihud, Y., & Mendelson, H. (1986). Asset pricing and the bid-ask spread. *Journal of Financial Economics*, *17*(2), 223–249.

Aquilina, M., Ibikunle, G., Rzayev, K., & Wang, X. (2025). High-frequency trading, the cost of capital, and market microstructure. *BIS Working Paper 1290*.

Azzimonti, M., & Fernandes, M. (2023). Social media networks, fake news, and polarization. *European Journal of Political Economy*, *76*, 102256.

Baker, S. R., Bloom, N., & Davis, S. J. (2016). Measuring economic policy uncertainty. *The Quarterly Journal of Economics*, *131*(4), 1593–1636. doi:10.1093/qje/qjw024

Barber, B. M., & Odean, T. (2008). All that glitters: The effect of attention and news on the buying behavior of individual and institutional investors. *Review of Financial Studies*, *21*(2), 785–818. doi:10.1093/rfs/hhm079

Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, *2008*(10), P10008. doi:10.1088/1742-5468/2008/10/P10008

Bodansky, D., Brunnée, J., & Rajamani, L. (2016). *International climate change law*. Oxford University Press.

Bolton, P., & Kacperczyk, M. (2021). Do investors care about carbon risk? *Journal of Financial Economics*, *142*(2), 517–549.

Bordalo, P., Gennaioli, N., & Shleifer, A. (2020). Memory, attention, and choice*. *The Quarterly Journal of Economics*, *135*(3), 1399–1442.

Bursztyn, L., Rao, A., Roth, C., & Yanagizawa-Drott, D. (2020). Misinformation during a pandemic. *NBER Working Paper: 27417*.

Bushman, R., & Smith, A. (2001). Financial accounting information and corporate governance. *Journal of Accounting and Economics*, *32*(1-3), 237–333.

Büthe, T., & Mattli, W. (2011). *The new global rulers: The privatization of regulation in the world economy*. Princeton, NJ: Princeton University Press.

Cameron, A. C., & Miller, D. L. (2015). A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, *50*(2), 317–372. doi:10.3368/jhr.50.2.317

Chordia, T., Roll, R., & Subrahmanyam, A. (2000). Commonality in liquidity. *Journal of Financial Economics*, *56*(1), 3–28.

Christensen, G., & Miguel, E. (2018). Transparency, reproducibility, and the credibility of economics research. *Journal of Economic Literature*, *56*(3), 920–980. doi:10.1257/jel.20171350

Cinelli, M., De Francisci Morales, G., Galeazzi, A., Quattrociocchi, W., & Starnini, M. (2021). The echo chamber effect on social media. *Proceedings of the National Academy of Sciences*, *118*(9), e2023301118.

Cobb, C. W. (2025). The politics of climate denialism and the secondary denialism of economics. *American Journal of Economics and Sociology*, *84*(1), 69–88. doi:10.1111/ajes.12606
Cologna, V., Baumberger, C., Knutti, R., Oreskes, N., & Berthold, A. (2022). The communication of value judgements and its effects on climate scientists' perceived trustworthiness. *Environmental Communication*, *16*(8), 1094–1107.
Cook, J., Ellerton, P., & Kinkead, D. (2018). Deconstructing climate misinformation to identify reasoning errors. *Environmental Research Letters*, *13*(2), 024018. doi:10.1088/1748-9326/aaa49f
De Long, J. B., Shleifer, A., Summers, L. H., & Waldmann, R. J. (1990). Noise trader risk in financial markets. *Journal of Political Economy*, *98*(4), 703–738.
Dees, S., & Ozyurt-Miller, S. (2026). *Climate economics and finance*. doi:10.1017/9781009628617
Delmas, M. A., & Burbano, V. C. (2011). The drivers of greenwashing. *California Management Review*, *54*(1), 64–87.
Diamond, D. W., & Verrecchia, R. E. (1991). Disclosure, liquidity, and the cost of capital. *Journal of Finance*, *46*(4), 1325–1360.
Dimson, E. (1979). Risk measurement when shares are subject to infrequent trading. *Journal of Financial Economics*, *7*(2), 197–226.
Drahos, N. (2025). Breaking ranks: Business fragmentation over climate policy reforms. *Business and Politics*, *27*(3), 442–464.
Driscoll, J. C., & Kraay, A. C. (1998). Consistent covariance matrix estimation with spatially dependent panel data. *The Review of Economics and Statistics*, *80*(4), 549–560.
Dye, R. A. (1985). Disclosure of nonproprietary information. *Journal of Accounting Research*, *23*(1), 123–145.
Easley, D., & O'Hara, M. (1992). Time and the process of security price adjustment. *The Journal of Finance*, *47*(2), 577–605.
Easley, D., & O'Hara, M. (1996). Liquidity, trading, and asset prices. *Journal of Financial Economics*, *41*(2), 261–290.
Engle, R. F., Giglio, S., Kelly, B., Lee, H., & Stroebel, J. (2020). Hedging climate change news. *Review of Financial Studies*, *33*(3), 1184–1216.
Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, *33*(1), 3–56.
Fama, E. F., & French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, *116*(1), 1–22.
Farrell, J. (2015). Corporate funding and ideological polarization about climate change. *Proceedings of the National Academy of Sciences*, *113*(1), 92–97.
Farrell, J. (2016). Network structure and influence of the climate change counter-movement. *Nature Climate Change*, *6*(4), 370–374.
Farrell, J., McConnell, K., & Brulle, R. (2019). Evidence-based strategies to combat scientific misinformation. *Nature Climate Change*, *9*, 191–195.
Federal Reserve Bank of St. Louis. (2025). FRED Economic Data. Accessed: 2025. Retrieved from https://fred.stlouisfed.org/
Flammer, C. (2021). Corporate green bonds. *Journal of Financial Economics*, *142*(2), 499–516.
Flynn, D., Nyhan, B., & Reifler, J. (2017). The nature and origins of misperceptions: Understanding false and unsupported beliefs about politics. *Political Psychology*, *38*(S1), 127–150.
Friede, G., Busch, T., & Bassen, A. (2015). Esg and financial performance: Aggregated evidence from more than 2000 empirical studies. *Journal of Sustainable Finance & Investment*, *5*(4), 210–233.
Gilardi, F., Alizadeh, M., & Kubli, M. (2023). Chatgpt outperforms crowd workers for text annotation tasks. *Proceedings of the National Academy of Sciences*, *120*(7), e2218185120.
Glosten, L. R., & Milgrom, P. R. (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, *14*(1), 71–100.
Goldstein, I., & Yang, L. (2017). Information disclosure in financial markets. *Annual Review of Financial Economics*, *9*, 101–125.
Grimmer, J., Roberts, M. E., & Stewart, B. M. (2022). *Text as data: A new framework for machine learning and the social sciences*. Princeton University Press.
Grossman, S. J., & Stiglitz, J. E. (1980a). On the impossibility of informationally efficient markets. *The American Economic Review*, *70*(3), 393–408.
Grossman, S. J., & Stiglitz, J. E. (1980b). On the impossibility of informationally efficient markets. *American Economic Review*, *70*(3), 393–408.
Hale, T., & Held, D. (2018). Breaking the cycle of gridlock. *Global Policy*, *9*(1), 129–137.

Han, B., Hirshleifer, D., & Walden, J. (2022). Social transmission bias and investor behavior. *Journal of Financial and Quantitative Analysis*, *57*(1), 390–412. doi:10.1017/S0022109021000077

Hasbrouck, J. (1991). Measuring the information content of stock trades. *The Journal of Finance*, *46*(1), 179–207.

Healy, P. M., & Palepu, K. G. (2001). Information asymmetry, corporate disclosure, and the capital markets: A review of the empirical disclosure literature. *Journal of Accounting and Economics*, *31*(1-3), 405–440.

Herzing, T. J., & Mück, M. (2025). Can misinformation be tolerated? *International Review of Financial Analysis*, *109*, 104768. doi:10.1016/j.irfa.2025.104768

Hong, H., Li, F. W., & Xu, J. (2019). Climate risks and market efficiency. *Journal of Econometrics*, *208*(1), 265–281. doi:10.1016/j.jeconom.2018.09.015

Horne, B., Nevo, D., & Smith, S. (2023). Ethical and safety considerations in automated fake news detection. *Behaviour & Information Technology*, *0*(0), 1–22.

Karpoff, J. M. (1987). The relation between price changes and trading volume: A survey. *Journal of Financial and Quantitative Analysis*, *22*(1), 109–126. doi:10.2307/2330874

Keohane, R. O., & Victor, D. G. (2011). The regime complex for climate change. *Perspectives on Politics*, *9*(1), 7–23.

Koreh, M., & Mandelkern, R. (2024). Trade-offs of government credibility institutions: Market credibility vs. social credibility. *Journal of European Public Policy*, *31*(11), 3539–3560.

Krueger, P., Sautner, Z., Tang, D., & Zhong, R. (2024). The effects of mandatory esg disclosure around the world. *Journal of Accounting Research*, *62*(5), 1795–1847.

Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, *53*(6), 1315–1335.

Lazer, D. M., Baum, M., Benkler, Y., et al. (2018). The science of fake news. *Science*, *359*(6380), 1094–1096.

Lei, L., & Luo, Y. (2023). Political/policy uncertainty, corporate disclosure, and information asymmetry. *Accounting Perspectives*, *22*(1), 87–110.

Leippold, M., Wang, Q., & Zhou, W. (2022). Machine learning in the chinese stock market. *Journal of Financial Economics*, *145*(2, Part A), 64–82.

Leuz, C., & Verrecchia, R. E. (2000). The economic consequences of increased disclosure. *Journal of Accounting Research*, *38*, 91–124.

Lewandowsky, S., Ecker, U. K., & Cook, J. (2017). Beyond misinformation: Understanding and coping with the "post-truth"' era. *Journal of Applied Research in Memory and Cognition*, *6*(4), 353–369.

Lins, K. V., Servaes, H., & Tamayo, A. (2017). Social capital, trust, and firm performance: The value of corporate social responsibility during the financial crisis. *Journal of Finance*, *72*(4), 1785–1824.

MacKinnon, J. G., Nielsen, M. Ø., & Webb, M. D. (2023). Cluster-robust inference: A guide to empirical practice. *Journal of Econometrics*, *232*(2), 272–299. doi:10.1016/j.jeconom.2022.04.001

Marquis, C., Toffel, M. W., & Zhou, Y. (2016). Scrutiny, norms, and selective disclosure: A global study of greenwashing. *Organization Science*, *27*(2), 483–504.

Mo, C., Hu, X., Imran, M., Khan, M. K., Guang, Y., & Xu, Z. (2026). Striking a balance: The crucial role of climate risk disclosure in correcting overvaluation and undervaluation in stock markets. *Financial Innovation*, *12*(102). doi:10.1186/s40854-025-00811-x

Moulton, B. R. (1990). An illustration of a pitfall in estimating the effects of aggregate variables on micro units. *The Review of Economics and Statistics*, *72*(2), 334–338.

Nyhan, B., & Reifler, J. (2010). When corrections fail: The persistence of political misperceptions. *Political Behavior*, *32*(2), 303–330.

Oreskes, N., & Conway, E. M. (2010). *Merchants of doubt: How a handful of scientists obscured the truth on issues from tobacco smoke to global warming*. New York: Bloomsbury Press.

Pankratz, S., Bauer, R., & Derwall, J. (2023). Climate change, firm performance, and investor surprises. *Management Science*, *69*(12), 7352–7398.

Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *Journal of Business*, *53*(1), 61–65. doi:10.1086/296071

Pastor, L., & Veronesi, P. (2012). Uncertainty about government policy and stock prices. *Journal of Finance*, *67*(4), 1219–1264.

Pástor, L., & Veronesi, P. (2013). Political uncertainty and risk premia. *Journal of Financial Economics*, *110*(3), 520–545.

Peng, L., & Xiong, W. (2006). Investor attention, overconfidence and category learning. *Journal of Financial Economics*, *80*(3), 563–602.

Petersen, M. A. (2009). Estimating standard errors in finance panel data sets: Comparing approaches. *The Review of Financial Studies*, *22*(1), 435–480. doi:10.1093/rfs/hhn053

Qiu, S., Liu, Q., Zhou, S., & Huang, W. (2022). Adversarial attack and defense technologies in natural language processing: A survey. *Neurocomputing*, *492*, 278–307.

Rajamani, L., & Brunnée, J. (2017). The legality of downgrading nationally determined contributions under the paris agreement: Lessons from the us disengagement. *Journal of Environmental Law*, *29*(3), 537–551.

Santurkar, S., Durmus, E., Ladhak, F., et al. (2023). Whose opinions do language models reflect? *arXiv preprint arXiv:2303.17548*.

Scholes, M., & Williams, J. (1977). Estimating betas from nonsynchronous data. *Journal of Financial Economics*, *5*(3), 309–327.

Sims, C. A. (2003). Implications of rational inattention. *Journal of Monetary Economics*, *50*(3), 665–690.

Stiglitz, J., & Kosenko, A. (2024a). The economics of information in a world of disinformation: A survey part 1: Indirect communication. *NBER Working Paper No. 32049*, 1–50.

Stiglitz, J., & Kosenko, A. (2024b). The economics of information in a world of disinformation: A survey part 2: Direct communication. *NBER Working Paper No. 32050*, 1–43.

Stiglitz, J. E. (2000). The contributions of the economics of information to twentieth century economics. *The Quarterly Journal of Economics*, *115*(4), 1441–1478.

Stock, J., & Watson, M. (2008). Heteroskedasticity-robust standard errors for fixed effects panel data regression. *Econometrica*, *76*(1), 155–174.

Stock, J. H., & Watson, M. W. (2002). Macroeconomic forecasting using diffusion indexes. *Journal of Business & Economic Statistics*, *20*(2), 147–162. doi:10.1198/073500102317351921

Stoll, H. R. (2000). Presidential address: Friction. *The Journal of Finance*, *55*(4), 1479–1514.

Supran, G., Rahmstorf, S., & Oreskes, N. (2023). Assessing exxonmobil's global warming projections. *Science*, *379*(6628), eabk0063.

Taber, C. S., & Lodge, M. (2006). Motivated skepticism in the evaluation of political beliefs. *American Journal of Political Science*, *50*(3), 755–769.

Tse, W. (2024). High-frequency trading, asset pricing, and market microstructure. *SSRN Working Paper*.

Verrecchia, R. E. (2001). Essays on disclosure. *Journal of Accounting and Economics*, *32*(1-3), 97–180.

Vosoughi, S., Roy, D., & Aral, S. (2018). The spread of true and false news online. *Science*, *359*(6380), 1146–1151.

Walter, N., Cohen, J., Holbert, R. L., & Morag, Y. (2020). Evaluating the impact of attempts to correct health misinformation on social media: A meta-analysis. *Health Communication*, *36*(13), 1776–1784. doi:10.1080/10410236.2019.1652634

Wardle, C., & Derakhshan, H. (2017). Information disorder: Toward an interdisciplinary framework for research and policy making. Council of Europe report DGI(2017)09.

Westwood, S. J., Grimmer, J., & Hall, A. B. (2025). *Measuring perceived slant in large language models through user evaluations* (Working Paper No. 4262). Stanford Graduate School of Business Working Paper No. 4262. Retrieved from https://www.gsb.stanford.edu/faculty-research/working-papers/measuring-perceived-slant-large-language-models-through-user

Williams, H. T. P., McMurray, J. R., Kurz, T., & Lambert, F. H. (2015). Network analysis reveals open forums and echo chambers in social media discussions of climate change. *Global Environmental Change*, *32*, 126–138. doi:10.1016/j.gloenvcha.2015.03.006

World Bank. (2025). World Bank Open Data. Accessed: 2025. Retrieved from https://data.worldbank.org/

Appendix

Section A reports descriptive statistics for the financial panel (Tables 6–11), and Section B reports coverage and composition for the climate-transition equity panel. Section C documents the misinformation corpus (Tables 18–22), including misinformation type and reach in Section C.1. Section D maps the dissemination network, Section E describes the exposure index and rubric-development checks, and Section F reports the heterogeneity and interaction checks.

A Financial Panel Descriptive Statistics

The merged firm-level panel comprises 19,183,738 firm-day observations across 5,375 unique equity securities spanning 2012–2025. Table 6 reports summary statistics - including observation counts, fill rates, means, standard deviations, and full distributional percentiles - for nine firm-level trading variables (prices, volume, market capitalization, dividend yield, bid-ask spreads, returns, and CAPM beta), three macroeconomic controls (GDP growth, unemployment, CPI inflation), and the five Fama-French factors (MktRF, SMB, HML, RMW, CMA). The table is presented in landscape orientation to accommodate the eleven-column layout.

The supplied equity assembly uses five stored region groups. These are input labels, not a validated listing-market classification; the spread analysis uses the exchange-based crosswalk instead. Table 7 reports the number of firm-day observations and unique RICs per region, alongside fill rates for closing prices, CAPM beta, and Fama-French factors. The stored CN group contributes 41.2% of input observations, the US group 31.7% and the JP group 23.8%; UK and EU groups account for the remainder. These percentages describe the supplied grouping, not corrected listing-market shares.

Temporal stability is a desirable property for panel econometric analysis, as changing panel composition can introduce survivorship bias. Table 8 reports annual observation counts, the number of distinct securities, and non-missing rates for key variables across all fourteen sample years. The constant RIC count of 5,375 per year describes the supplied calendar grid, rather than demonstrating a survivor-bias-free investment universe, while data fill rates for prices, returns, and betas increase monotonically over time as coverage improves.

Before estimating the main regression specifications, it is useful to examine the bivariate relationships among key continuous variables for evidence of multicollinearity or unexpected sign patterns. Table 9 presents the upper-triangle pairwise correlation matrix for daily returns, CAPM beta, percentage bid–ask spread, volume, market capitalization, and the three primary Fama-French factors. Correlations are generally modest in magnitude, with the largest off-diagonal entries appearing between MktRF and SMB ($r = -0.14$) and between MktRF and HML ($r = -0.15$).

Fama-French factor returns are matched to each firm based on its listing region, with four regional factor sets covering Asia Pacific, the United States, Japan, and Europe. Table 10 reports the number of matched observations, fill rates, and mean daily values for all five factors by region. The US market excess return exhibits the highest mean (0.056% per day), while Japan exhibits the largest standard deviation (1.153%).

The microstructure variables reported in Table 11 include the percent quoted spread used as the primary liquidity outcome and supplied daily price-gap fields. The table documents observation counts, the number of securities with non-missing data, and distributional summaries for eight trade-level variables: bid–ask spread, percentage spread, order flow, trade intensity, supplied buyer/seller indicators, volume-weighted average price (VWAP), and the absolute VWAP–midpoint percentage gap. Source field names do not identify signed order-flow quantities or per-dollar transaction impact. Coverage is near-universal for spread and order-flow variables (5,343 of 5,375 RICs) but lower for trade intensity (2,029 RICs), reflecting cross-exchange differences in tick-level data availability.

Variable	Unit	N	Fill%	Mean	SD	Min	P25	Median	P75	Max
Firm-Level Variables										
Price Close	source currency	13,759,934	71.7	388.67	2,128.98	0.01	10.96	30.58	118.26	359,900.00
Volume	shares	18,394,237	95.9	9.03M	60.58M	0.00	0.00	0.823M	5.08M	45,941.76M
Company Market Cap	source currency	14,263,811	74.4	1,860,901.54	21,101,237.64	14.87	5,032.21	15,080.74	104,672.42	2,464,175,093.59
Dividend yield	%	12,626,836	65.8	2.38	8.49	0.00	0.48	1.72	3.49	20,316.87
BidAskSpread	source currency	13,419,146	70.0	4.24	61.02	-268.00	0.01	0.04	0.30	13,550.00
SpreadPercent	%	13,419,145	70.0	0.54	1.95	-200.00	0.07	0.16	0.37	172.14
DailyReturn		13,255,243	69.1	0.09	115.33	-73.29	-1.04	0.00	1.01	419,653.09
CAPM_Beta		13,802,427	71.9	0.54	0.48	-11.74	0.22	0.47	0.82	298.91
TotalReturn		13,735,222	71.6	0.10	113.30	-98.58	-1.04	0.00	1.02	419,653.09
Macroeconomic Controls										
GDP Growth	%	19,183,738	100.0	3.55	2.90	-10.05	1.82	2.79	6.07	8.57
Stored unemployment series	%	19,183,738	100.0	74.87	12.97	53.10	63.52	80.07	82.02	92.19
CPI Inflation	%	19,183,738	100.0	1.89	1.52	-0.23	0.98	1.92	2.62	8.83
Fama–French Five Factors										
FF MktRF	%	18,987,538	99.0	0.03	1.01	-12.01	-0.44	0.05	0.55	10.03
FF SMB	%	18,987,538	99.0	-0.01	0.53	-4.81	-0.30	0.00	0.29	5.71
FF HML	%	18,987,538	99.0	0.01	0.66	-5.03	-0.32	0.00	0.33	6.73
FF RMW	%	18,987,538	99.0	0.01	0.43	-3.76	-0.23	0.00	0.25	4.25
FF CMA	%	18,987,538	99.0	0.00	0.40	-2.92	-0.21	0.00	0.21	2.47

Table 6: Panel A: Summary statistics for key variables in the merged dataset. The sample spans 2012–2025 and covers 5,375 RICs. Fill% indicates the proportion of non-missing observations for each variable. All firm-level variables are at daily frequency.

Region	Obs	Share%	RICs	First	Last	Price%	Beta%	FF%
CN	7,897,640	41.2	2,231	2012-01-02	2025-12-31	62.3	63.0	100.0
US	6,075,274	31.7	1,712	2012-01-02	2025-12-31	71.6	69.7	97.0
JP	4,558,053	23.8	1,253	2012-01-02	2025-12-31	85.2	87.4	99.7
UK	364,446	1.9	100	2012-01-02	2025-12-31	93.4	92.7	100.0
EU	288,325	1.5	79	2012-01-02	2025-12-31	94.2	92.8	100.0
Total	**19,183,738**	100.0	**5,375**					

Table 7: Panel B: Coverage by stored region group. Share% is the fraction of total firm-day observations. Price%, Beta%, and FF% indicate the fill rate for Price Close, CAPM Beta, and Fama–French factors, respectively.

Year	Obs	RICs	Price%	Return%	Beta%
2012	1,363,092	5,375	53.6	51.3	25.2
2013	1,356,566	5,375	54.7	52.2	57.3
2014	1,366,584	5,375	55.8	53.3	58.0
2015	1,365,143	5,375	56.9	54.4	59.4
2016	1,367,945	5,375	59.3	56.5	61.2
2017	1,360,941	5,375	64.4	61.8	64.9
2018	1,368,416	5,375	68.1	65.4	70.3
2019	1,373,060	5,375	71.3	68.5	73.4
2020	1,376,595	5,375	75.4	72.7	76.8
2021	1,377,392	5,375	82.4	80.0	83.1
2022	1,373,896	5,375	87.2	84.6	89.3
2023	1,372,026	5,375	90.1	87.4	93.1
2024	1,382,316	5,375	91.8	88.8	96.5
2025	1,379,766	5,375	92.4	89.6	97.7

Table 8: Panel C: Annual coverage. Obs is the number of firm-day observations per year; RICs is the number of distinct securities observed that year. Price%, Return%, and Beta% are non-missing rates for Price Close, DailyReturn, and CAPM Beta.

	Return	Beta	Spread%	Volume	MktCap	MktRF	SMB	HML
Return	1.00	0.17	0.02	0.00	0.03	0.00	-0.00	-0.00
Beta		1.00	-0.23	-0.01	-0.05	-0.00	-0.00	-0.00
Spread%			1.00	-0.02	0.01	0.00	-0.00	-0.01
Volume				1.00	0.07	-0.00	0.00	0.00
MktCap					1.00	0.00	-0.00	-0.00
MktRF						1.00	-0.14	-0.15
SMB							1.00	0.05
HML								1.00

Table 9: Panel D: Pairwise correlations among key variables. Empty cells below the diagonal are suppressed for readability.

FF Region	Obs	Fill%	MktRF	SD(MktRF)	SMB	HML	RMW	CMA
AsiaPac	7,897,640	100.0	0.019	0.856	-0.007	0.021	0.018	0.006
US	6,075,274	97.0	0.056	1.084	-0.009	-0.003	0.006	-0.003
Japan	4,558,053	99.7	0.033	1.153	0.002	0.012	0.000	0.005
Europe	652,771	100.0	0.033	1.028	0.000	0.009	0.006	0.001

Table 10: Panel E: Fama-French five-factor summary by region mapping. Mean daily factor returns are reported. Fill% indicates the proportion of firm-day observations matched to factors.

Variable	N	RICs	Mean	SD	Median
BidAskSpread	13,419,146	5,343	4.24	61.02	0.04
SpreadPercent	13,419,145	5,343	0.54	1.95	0.16
OrderFlow	13,268,923	5,343	0.08	193.08	0.00
TradeIntensity	5,146,749	2,029	4,750.04	12,478.89	2,198.00
BuyerInitiated	13,837,387	5,343	0.48	0.50	0.00
SellerInitiated	13,837,387	5,343	0.47	0.50	0.00
VWAP	13,584,674	5,343	392.92	2,115.66	30.53
PriceImpact	13,268,922	5,343	0.68	47.95	0.37

Table 11: Panel F: Microstructure-variable coverage. The percent quoted spread is the main liquidity outcome. PriceImpact is a supplied field name for the absolute VWAP–midpoint percentage gap, not transaction price impact. N is the number of non-missing observations; RICs is the number of distinct securities with at least one non-missing value.

Table 12: Mean CAPM Beta by Region and Market-Capitalisation Quintile

Region	Mean β	Median β	Q1 (Small)	Q2	Q3	Q5 (Large)
CN	0.655	0.594	0.643	0.673	0.667	0.600
US	0.488	0.388	0.806	0.837	0.823	0.166
JP	0.415	0.330	0.263	0.283	0.303	0.825
UK	0.747	0.714	0.731	0.785	0.782	0.570
EU	0.905	0.877	0.816	0.967	0.897	0.995

Notes: Mean rolling CAPM beta computed from 252-day OLS windows requiring minimum 126 observations. Quintiles defined by Company Market Cap within each region-day. The low overall mean (0.54) is driven by Japanese firms (0.42) and the preponderance of small-cap US firms in the sample. Non-synchronous trading bias (Scholes and Williams, 1977) likely contributes to downward bias in thinly traded securities.

B Climate-Transition Panel Descriptive Statistics

This subsection reports descriptive statistics for the unique climate-transition firm panel used in the subsample regressions. The objective is to document coverage, composition, and distributional properties before estimation. Across all three tables below, the sample is defined by renewable and fossil-fuel TRBC mappings with utilities excluded, consistent with the main-text climate-transition specifications.

Table 13: Climate-Transition Panel Coverage Summary

Metric	Value
Firm-day observations	726,890
Unique securities (RICs)	204
Start date	2012-01-02
End date	2025-12-31
Unique year-month bins	168
Median observations per firm	3,588

Notes: Coverage statistics for the full financial input, before the content-window, quote, trading and complete-case filters. These are not regression sample sizes.

Table 14: Climate-Transition Panel Composition by Stored Region Group (Unique Securities)

Region	Unique securities	Share (%)
US	102	50.00
CN	83	40.69
JP	15	7.35
EU	2	0.98
UK	2	0.98

Notes: Shares describe the supplied region groups for 204 securities, not actual listing markets. The market crosswalk and corrected market composition are reported separately.

Table 15: Climate-Transition Panel Key Variable Descriptive Statistics

Variable	N	Mean	Std. Dev.	P25	Median	P75
SpreadPercent	504,043	0.6147	1.6705	0.0607	0.1590	0.4090
CAPM_Beta	525,697	0.5679	0.4984	0.2039	0.5168	0.8864
ExcessReturn	506,612	0.0328	2.6949	-1.2207	-0.0100	1.1272
log_volume	698,179	11.3694	6.8472	5.7071	14.5147	16.3631
log_mktcap	559,387	25.0607	2.7631	22.9046	24.3402	27.1216
IntradayVolatility	496,614	3.4611	2.5748	1.7758	2.7528	4.3222
DivYield	464,457	2.9153	3.6049	0.2509	1.8536	4.1325
cpi_inflation	726,890	2.1371	1.5976	1.2616	2.0000	2.6195
gdp_growth	726,890	3.8414	2.6520	2.3402	2.8876	6.0685
unemployment (stored)	726,890	73.1123	11.1097	63.5234	80.0325	80.4977

Notes: Statistics correspond to the estimation sample after standard preprocessing used in the econometric pipeline (including transformations and winsorisation for relevant continuous variables).

The descriptive profile in Tables 13–15 motivates the energy-subsample strategy. The panel is sufficiently long and dense for two-way FE estimation, yet actual listing composition is concentrated in mainland China and Indonesia, making local market-day effects and inference at the sector assignment level important.

Table 16: Listing-market composition of the principal sample

Listing market	Securities	Security-days	Mean spread
Brazil	8	18,945	0.256
France	1	3,519	0.038
Germany	1	1,286	0.085
Hong Kong	5	16,465	0.226
Indonesia	62	111,633	1.756
Japan	3	10,104	0.172
Mainland China	82	197,427	0.157
Malaysia	1	3,373	0.505
Netherlands	1	3,516	0.042
Taiwan	12	32,073	0.622
United Kingdom	2	6,962	0.028
United States	24	80,497	0.070

Market comes from the listing-exchange suffix. Mean spreads are raw percentages before tail capping. Market-day effects absorb all observed local securities together; counts do not measure independent shocks.

Table 17: Estimation sample construction

Filter	Security-days
Financial input	726,890
Within content window	715,200
Positive valid quotes and active trading	489,929
Contemporaneous-control complete cases	486,655
Prior-session-control complete cases	485,800

The two complete-case rows are alternative control choices applied to the same eligible quote sample, not sequential filters. Content window is 3 January 2012 to 13 October 2025. Active trading requires positive volume and a positive close.

C Descriptive Statistics and Supporting Tables

The corpus comprises 265,652 public Facebook posts with platform-reported engagement data. Table 18 provides distributional summaries for all engagement metrics - likes, shares, comments, total reactions (including emoji-specific breakdowns for angry, love, haha, sad, wow, and care reactions), and view counts - as well as text-level characteristics (character length and word count). The large gap between mean and median values for most engagement variables reflects the heavy right skew characteristic of social media diffusion, where a small number of viral posts account for a disproportionate share of total engagement.

Table 18: Social Media Post Summary Statistics

Variable	N	Mean	Median	Std.Dev.	Min	Max
Likes	262,461	140.8	10.00	2,015.6	0.000	525,472.0
Shares	265,651	44.90	2.00	682.3	0.000	81,948.0
Comments	265,651	22.72	0.000	217.6	0.000	39,790.0
Total Reactions	262,492	175.9	13.00	2,278.2	0.000	531,508.0
Views	220,511	10,372.8	1,625.0	91,127.8	101.0	15,900,642
Angry Reactions	262,461	7.69	0.000	225.5	0.000	57,470.0
Love Reactions	262,461	15.38	0.000	267.5	0.000	46,490.0
Haha Reactions	262,461	3.84	0.000	89.18	0.000	31,575.0
Sad Reactions	262,461	3.52	0.000	162.2	0.000	44,420.0
Wow Reactions	262,461	1.66	0.000	31.79	0.000	7,074.0
Care Reactions	262,461	0.000	0.000	0.000	0.000	0.000
Text Length (chars)	265,652	3,599.1	1,337.0	6,828.6	2.00	168,477.0
Word Count	265,652	553.9	197.0	1,062.6	1.00	26,148.0

Notes: Summary statistics for 265,652 social media posts from the Meta Content Library (MCL). Engagement metrics reflect platform-reported counts at the time of data collection.

The 217,954 unique posts paired by calendar date to the climate-transition panel span 2012 through

2025. Table 19 reports their annual distribution, with the misinformation classification produced by the Qwen3.6-35B-A3B model (Section 4.3): a post is counted as misinformation if its continuous misinformation score reaches the rubric's "clearly false or pseudoscientific climate/energy claims" threshold of 50 on the 0–100 scale. Post volume rises sharply after 2019, coinciding with the intensification of global climate activism and the digital migration associated with the COVID-19 pandemic, while Qwen-classified misinformation prevalence fluctuates between 1.7% and 5.0% across years. The within-year movement is not proportional to raw post volume: prevalence peaks in the contested-policy years 2012 (5.0%), 2017 (3.0%), 2019 (3.2%), and 2022 (3.0%), and falls to its sample minimum in the highest-volume years 2024–2025 (≈1.7%), where general climate content grows faster than misinformation.

Table 19: Annual Distribution of Posts and Qwen-Classified Misinformation Prevalence

Year	Total Posts	Misinfo. Posts	Prevalence (%)	Mean Likes	Mean Shares
2012	1,276	64	5.02	86.2	38.96
2013	2,649	82	3.10	87.5	39.97
2014	3,885	88	2.27	87.8	33.56
2015	5,106	95	1.86	115.5	47.93
2016	9,954	272	2.73	138.2	82.97
2017	13,845	413	2.98	93.7	58.13
2018	11,429	268	2.34	66.2	45.01
2019	17,160	542	3.16	90.6	47.83
2020	17,118	427	2.49	128.0	61.19
2021	23,038	469	2.04	112.9	38.67
2022	24,102	710	2.95	86.9	22.68
2023	25,846	731	2.83	76.7	21.54
2024	30,404	538	1.77	88.3	26.95
2025	32,142	552	1.72	344.6	62.91
Total	217,954	5,251	2.41	132.4	43.03

Notes: Annual counts for the 217,954 unique posts paired by date to the climate-transition panel. "Misinfo. Posts" counts posts whose Qwen3.6-35B-A3B misinformation score reaches the threshold of 50; "Prevalence" is that count as a share of the year's posts.

The Meta Content Library organizes posts into eight keyword-based thematic datasets covering different dimensions of climate discourse. Table 20 breaks down post coverage, Qwen-classified misinformation counts, engagement, and unique owner counts by source dataset. The pattern is intuitive given how the model reasons about content: the explicitly political "combined" multi-keyword set (6.4% prevalence) and the climate-politics set (1.9%) concentrate the misinformation, because that is where contested claims about climate science and policy are made, whereas the adaptation, resilience, and ecological datasets—which collect operational and scientific content—sit at or below 0.2% prevalence. The classifier is not flagging climate *topicality*; it is flagging the false or misleading *claims* that cluster in the political corners of the corpus.

Table 20: Post Coverage by MCL Source Dataset (Qwen-Classified Misinformation)

Source Dataset	Posts	Misinfo.	Prev. (%)	Mean Likes	Mean Shares	Owners
Combined	65,688	4,180	6.4	144.8	69.73	25,171
Climate Change Adaptation Al..	61,079	126	0.2	48.10	11.12	17,931
Climate Change Politics All ..	32,930	618	1.9	153.2	69.05	14,527
Climate Change Global Justic..	14,406	134	0.9	120.4	38.57	7,369
Climate Change Global Resili..	13,448	18	0.1	68.60	11.55	7,019
Climate Change Activism All ..	11,222	149	1.3	639.8	62.98	6,120
Ecological Resilience All Time	9,835	8	0.1	77.90	16.27	5,710
Ecological Justice All Time	9,346	18	0.2	77.30	28.76	4,984

Notes: The 217,954 paired posts drawn from eight MCL keyword-based search datasets. "Combined" denotes the multi-keyword merged set. "Misinfo." counts posts with Qwen3.6-35B-A3B misinformation score ≥ 50; "Owners" is the count of distinct posting accounts per dataset.

Understanding who produces misinformation requires examining the distribution of posting activity across the 55,699 unique post owners in the paired corpus. Table 21 groups owners into five activity brackets and reports each bracket's contribution to total volume and Qwen-classified misinformation output. The distribution is heavily right-skewed: over half of all owners contribute only a single post, while 124 heavy posters (more than 100 posts each) account for roughly 12% of total volume. The Qwen misinformation rate is broadly stable across brackets (1.3%–2.6%), which carries a substantive point—misinformation in this corpus is not the product of a small set of hyperactive accounts but is distributed across the full activity range, including the long tail of one-time posters. The dissemination problem is therefore one of breadth rather than concentration in supply, even if reach (Section C.1) is far more skewed than production.

Table 21: Post Owner Activity and Qwen-Classified Misinformation Concentration

Activity Bracket	Owners	Total Posts	% of Corpus	Misinfo. Posts	Misinfo. Rate (%)
1 post	30,528	30,528	14.0	669	2.19
2–10 posts	21,627	78,690	36.1	2,000	2.54
11–50 posts	3,152	63,464	29.1	1,639	2.58
51–100 posts	268	18,280	8.4	233	1.27
>100 posts	124	26,990	12.4	710	2.63
Total	55,699	217,952	100.0	5,251	2.41

Notes: Owners of the 217,954 paired posts (two posts have no owner identifier) grouped by posting frequency. Misinformation rate is the share with Qwen3.6-35B-A3B misinformation score ≥ 50.

The corpus spans multiple content formats and platform surface types, which may shape how misinformation circulates. Table 22 reports the distribution of posts across three metadata dimensions: content type (photos, links, albums, status updates, videos, reshares, and others), surface type (page, profile, group, or event), and owner type (page or profile). Photos are the most common content format at 35.4%, and pages dominate both surface type (90.1%) and owner type (91.7%), reflecting the institutional character of climate-related Facebook activity.

Table 22: Post Metadata: Content Type, Surface Type, and Owner Type

Content	N	%	Surface	N	%	Owner	N	%
photos	94,095	35.4	page	239,340	90.1	Page	243,708	91.7
links	73,672	27.7	profile	19,724	7.4	Profile	21,944	8.3
albums	46,457	17.5	group	6,236	2.3			
status	26,390	9.9	event	352	0.1			
videos	16,524	6.2						
reshare	5,444	2.0						
others	15	0.0						

Notes: Distribution by content type, surface type, and owner type. Pages dominate (91.7%).

C.1 Misinformation Type and Reach in the Qwen Classification

The Qwen3.6-35B-A3B classifier returns more than a binary label. Alongside the continuous 0–100 misinformation score, it assigns each flagged post one of six type labels—denial, conspiracy, misleading framing, pseudoscience, greenwashing, or other—which makes the *kind* of climate misinformation in the corpus observable, not just its quantity. Table 23 reports the composition. Outright *denial* of climate science is the single largest category (49.8% of flagged posts), followed by *conspiracy* framings (30.8%) and *misleading framing* of otherwise-accurate facts (17.8%); pseudoscience and greenwashing are rare. The intuition the typology captures is that the misinformation reaching these markets is predominantly the hard-denial and conspiracy content that organized political campaigns produce, not the softer greenwashing that corporate communications produce—a distinction a binary classifier cannot draw.

Table 23: Composition of Qwen-Classified Climate Misinformation by Type

Type	Posts	% of Misinfo.	% of Corpus	Mean Likes	Mean Shares
Denial	2,615	49.8	1.200	226.1	129.6
Conspiracy	1,616	30.8	0.741	122.2	103.9
Misleading framing	937	17.8	0.430	181.2	115.2
Pseudoscience	81	1.5	0.037	290.1	42.9
Greenwashing	2	0.0	0.001	19.5	11.0
All misinformation	5,251	100.0	2.410	187.2	117.7

Notes: Posts with Qwen3.6-35B-A3B misinformation score ≥ 50, by the model's modal non-null type label. "% of Corpus" is the share of all 217,954 paired posts. Engagement is platform-reported at collection.

Two further facts about the supplied corpus summaries motivate distinguishing the amount of content from its engagement. The financial index itself is not engagement-weighted. First, misinformation travels further than other climate content: the 5,251 flagged posts average 187 likes and 118 shares against 131 likes and 41 shares for the non-flagged remainder, and their median share count is 11 against 2—a roughly fivefold gap at the median that is consistent with the broader finding that false content diffuses disproportionately (Vosoughi et al., 2018). Second, the within-type engagement ordering is itself informative: denial and conspiracy posts, the categories produced by politically motivated actors, are shared two-to-three times as often as the corpus average, whereas pseudoscience—which is read and liked but less often forwarded—has the highest mean likes but the lowest mean shares of any misinformation type. Reach, not production, is where the asymmetry lives.

Figure 3 plots monthly Qwen misinformation prevalence over the sample. Prevalence is visibly episodic rather than trending: it spikes around contested-policy episodes and recedes between them, and the post-2019 surge in raw post volume (vertical line) dilutes prevalence even as the absolute count of misinformation posts rises. Figure 4 decomposes the annual flagged count by type, showing that denial and conspiracy content drive both the level and the post-2019 growth of misinformation in the corpus, while misleading framing grows more slowly and pseudoscience stays negligible.

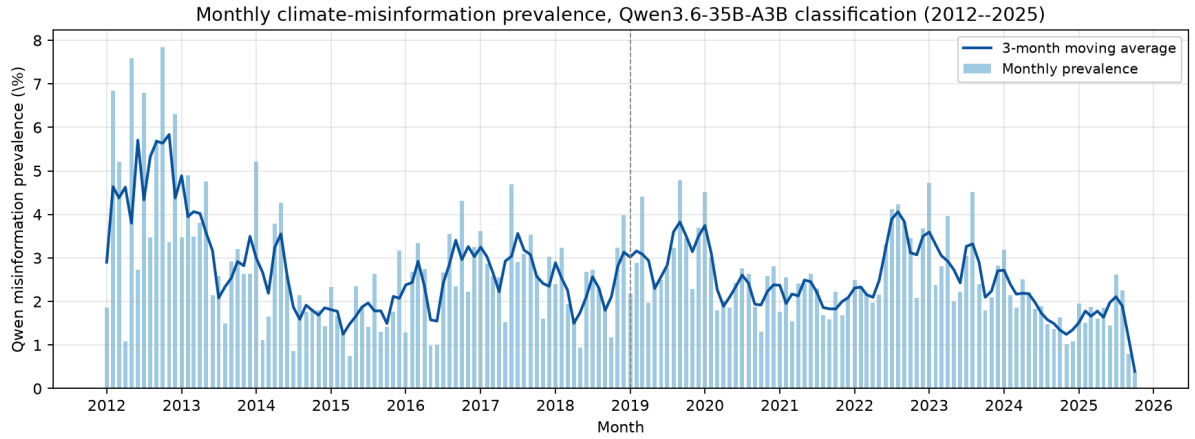


Figure 3: Monthly climate-misinformation prevalence in the Qwen3.6-35B-A3B classification, 2012–2025. Bars are monthly prevalence (flagged posts as a share of that month's paired posts); the solid line is a three-month moving average. The dashed line marks January 2019.

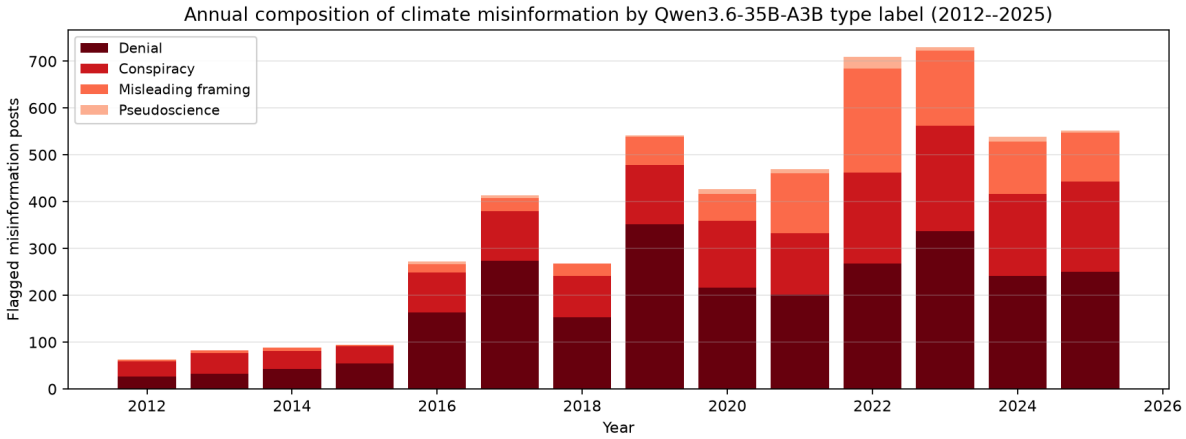


Figure 4: Annual composition of Qwen-classified climate misinformation by type, 2012–2025. Denial and conspiracy framings account for the bulk of flagged content and for its post-2019 growth.

D Network Structure of Misinformation Dissemination

I supplement the post-level corpus with an owner-based network constructed on the 150 most active and engaged post owners (pages and profiles), selected via a composite ranking weighting total engagement (40%), misinformation post count (40%), and overall posting activity (20%). Two owners are connected if their monthly posting distributions correlate at $\rho \geq 0.3$ (Pearson). The largest connected component contains 142 nodes and 1,004 edges, with five Louvain communities (Blondel, Guillaume, Lambiotte, and Lefebvre, 2008) of community misinformation rates varying from 4.2% to 21.7%. The threshold captures accounts that post more during the same months, which may reflect coordinated dissemination, shared thematic interests, or common responses to COP meetings, extreme weather, or elections; the network structure should therefore be read as thematic co-occurrence rather than necessarily coordinated inauthentic behaviour (Walter, Cohen, Holbert, and Morag, 2020; Williams, McMurray, Kurz, and Lambert, 2015).

Three centrality measures bear on the empirical specification that follows. Eigenvector centrality is negatively correlated with the per-owner misinformation rate ($r = -0.296$, $p = 0.0004$), and degree centrality is positively correlated with the absolute misinformation count ($r = 0.218$, $p = 0.009$); betweenness centrality shows no significant association ($r = -0.068$, $p = 0.42$). The structurally central nodes disseminate proportionally lower rates of misleading content, while peripheral high-degree nodes generate larger absolute volumes. This pattern weakens, rather than supports, the prior assumption that engagement-weighting captures "reach-amplified misinformation": if reach were dominated by misinformation-heavy actors, eigenvector centrality should covary positively, not negatively, with misinformation rate. Figures 5 and 6 display the network and centrality scatterplots respectively.

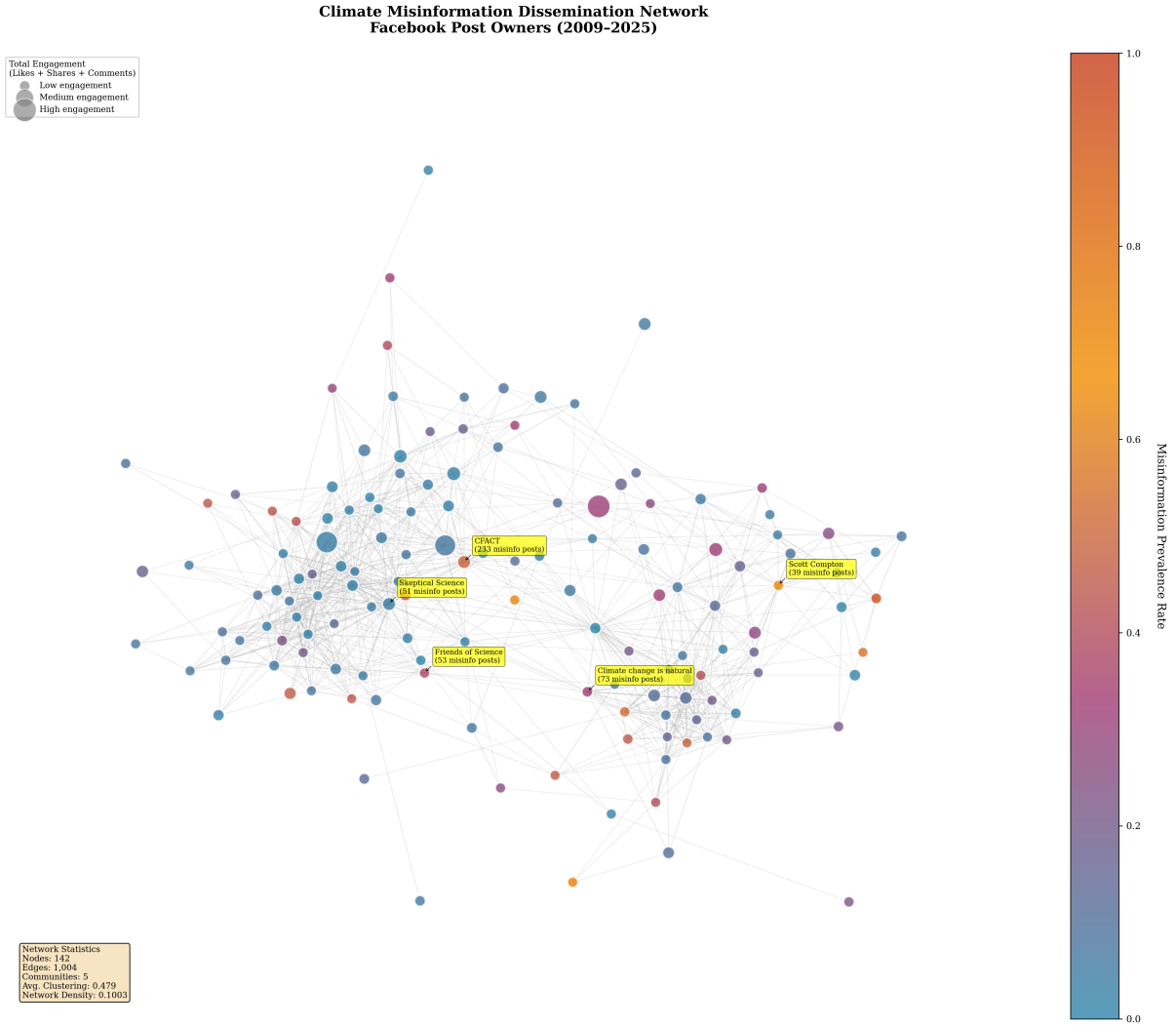


Figure 5: Climate Misinformation Dissemination Network

Notes: Network graph of 142 Facebook post owners (pages/profiles) constructed from temporal posting correlations (Pearson $\rho \geq 0.3$). Node size is proportional to total engagement (likes + shares + comments). Node color indicates misinformation prevalence rate (blue = low, red = high). Edge thickness reflects correlation strength. Top 5 misinformation spreaders are labeled. Force-directed layout (Fruchterman–Reingold). Network statistics: 5 communities (Louvain), average clustering = 0.48, density = 0.10. Data source: Meta Content Library, Dec 2009-Oct 2025.

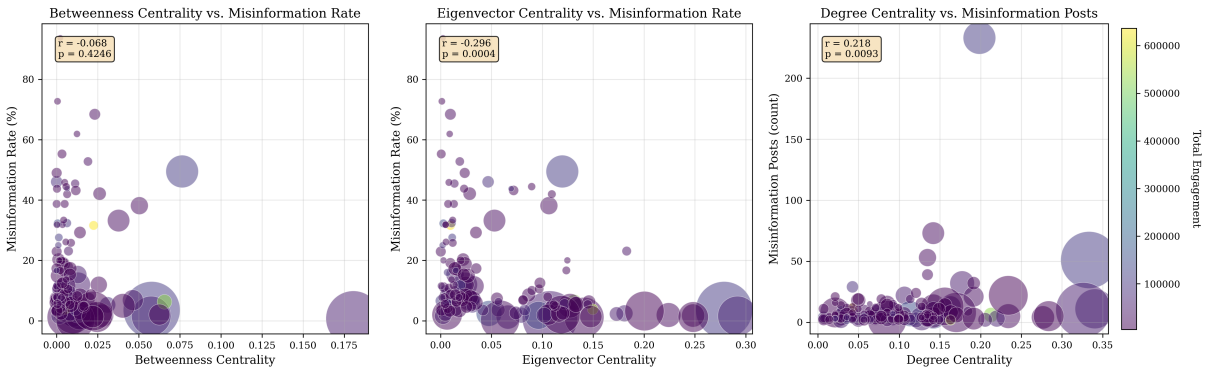


Figure 6: Network Centrality and Misinformation Prevalence

Notes: Scatterplots of three centrality measures against misinformation prevalence and count. **Left panel:** Betweenness centrality vs. misinformation rate ($r = -0.068$, $p = 0.42$). **Center panel:** Eigenvector centrality vs. misinformation rate ($r = -0.296$, $p = 0.0004$). **Right panel:** Degree centrality vs. misinformation post count ($r = 0.218$, $p = 0.009$). Node size represents total posts. Color indicates total engagement. Eigenvector centrality exhibits a statistically significant negative association with misinformation rate, suggesting influential nodes spread less misinformation.

E Firm-Day Exposure: Descriptive Statistics and Validation

This appendix reports the descriptive statistics of the firm-day misinformation-exposure index introduced in Section 4.3 and the rubric-development spot-check used to assess the classifier's handling of satire and politically loaded but non-climate content.

E.1 Descriptive Statistics

Table 24 summarises the firm-day index and its constituent scores across the 714,227 firm-days in the climate-transition panel with at least one paired post. The index $\text{Exp}_{i,t}^{\text{mean}}$ has a mean of 0.34 on a 0–100 scale, a median of 0.09, a 99th percentile of 2.73, and a share of exactly-zero firm-days of 44%. The sum variant $\text{Exp}_{i,t}^{\text{sum}}$ has mean 21.85 and 99th percentile 171.50; the log-transformed version $\log(1+\text{Exp}_{i,t}^{\text{sum}})$ is approximately normal in shape with mean 1.80 and 99th percentile 5.15. The misinformation sub-score $M_{i,t}^{\text{mean}}$ has mean 1.73, p99 15.60, and a share of exactly-zero firm-days of 41%, reflecting the rubric's restriction of misinformation scoring to climate-or-energy claims. The relevance sub-score $R_{i,t}^{\text{mean}}$ has mean 12.59 and p99 27.17, with only 2% of firm-days exactly at zero, indicating that climate posts paired by date are typically at least weakly relevant to the firm's sub-industry.

Table 24: Descriptive statistics of the firm-day misinformation-exposure index (climate-transition panel, 714,227 firm-days with ≥ 1 climate post)

Variable	Mean	SD	p_{50}	p_{90}	p_{99}	Share zero
# climate posts paired to the firm-day	59.900	49.449	52.000	127.000	208.000	0.017
Mean per-post sector relevance R^{mean}	12.594	5.106	12.231	18.524	27.173	0.022
Mean per-post climate misinformation M^{mean}	1.731	3.232	0.781	4.268	15.600	0.408
Firm-day mean exposure Exp^{mean}	0.338	0.715	0.090	0.971	2.732	0.442
Firm-day max exposure Exp^{max}	13.526	20.236	7.000	50.700	68.640	0.442
Firm-day sum exposure Exp^{sum}	21.846	37.661	7.500	71.250	171.500	0.442
Share of posts with $M \geq 25$	0.020	0.037	0.009	0.050	0.167	0.464
$\log(1+\text{Exp}^{\text{sum}})$	1.795	1.758	2.140	4.280	5.150	0.442

Notes: Sample is firm-days in the energy panel for which at least one English-language Facebook climate post on the same calendar date was scored against the firm's TRBC industry. "Share zero" is the fraction of firm-days for which the variable evaluates to exactly 0 (typically because no posts in the day's classification batch were judged sector-relevant or misinformation-laden).

Figure 7 plots the distribution of the headline index and its log-volume variant on non-zero firm-days. The mean variant has a long right tail truncated by the bound of the 0–100 scale; the log-volume variant is more symmetric, justifying its use in the variant Model I specification reported in Table 1.

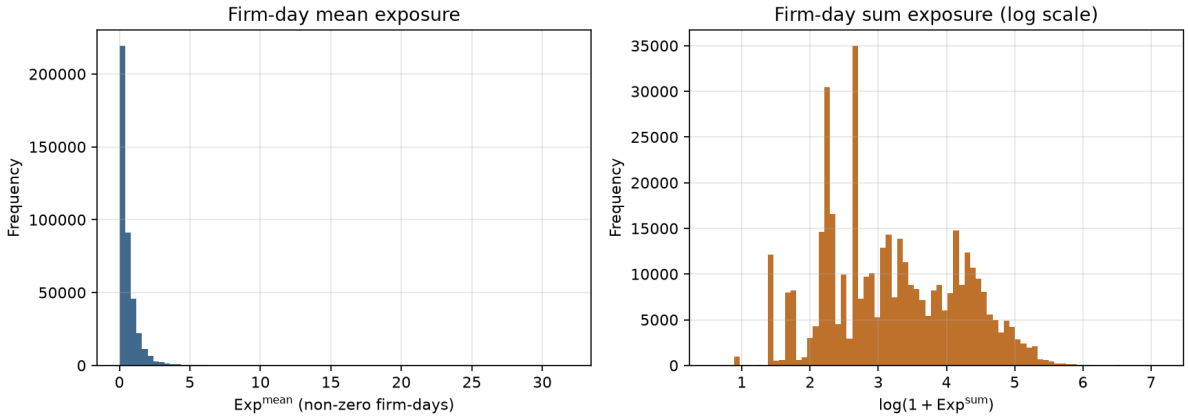


Figure 7: Firm-day exposure distribution. Left panel: $\text{Exp}_{i,t}^{\text{mean}}$ on non-zero firm-days. Right panel: $\log(1+\text{Exp}_{i,t}^{\text{sum}})$.

E.2 Classifier Validation

The classifier was assessed during rubric development using a spot-check of the first 2,000 (post, sub-industry) judgments. Two qualitative findings shaped the final rubric. First, a default rubric ("score climate misinformation on a continuous 0–100 scale") over-scored politically charged but non-climate content. Anti-vaccine conspiracy posts, antisemitic content about EU politicians, and electoral-fraud posts that happened to be paired to climate sub-industries by calendar date received high misinformation scores under the default rubric despite making no empirical claims about climate or energy. The headline composite $\text{Exp}_{p,s} = R_{p,s} \cdot M_{p,s}/100$ partially neutralised this leakage because relevance scores on sector-irrelevant content were low, but the misinformation sub-score itself was contaminated. The final rubric (Section 4.3) explicitly restricts misinformation scoring to climate-or-energy claims, and a re-run of the spot-check under the final rubric produced misinformation scores of zero on the politically charged but non-climate items.

Second, sarcastic and parodic content that mocked climate denial received high misinformation scores under the default rubric. A representative example begins "MDPI, the predatory journal of which nothing can be believed, according to Facebook climate experts, recently published a dubious study on anthropogenic warming…" and ends with the parody credit "A DONT BELIEVE EVERYTHING YOU READ PRODUCTION". The default rubric scored this post at $M_{p,s} = 95$ with the type label "denial"; the final rubric, which includes an explicit instruction to treat satire and parody as not-misinformation, scores it at $M_{p,s} = 0$ with the rationale "satirical parody mocking climate denialism, not a sincere assertion of false claims". Both rubric refinements were committed to the spec before the production run, and the production run used the final rubric.

E.3 Within-Classifier Relevance Checks

The natural relevance check compares highly sector-relevant posts with those assigned a low relevance score. A high-relevance conditional mean and its contribution to the all-post mean are different regressors. If $H_p = 1\{R_{p,s} \geq c\}$, q^H is the qualifying-post share and $\bar{e}^H$ its mean exposure, then $C^H = q^H \bar{e}^H$ and $E = C^H + C^L$. This decomposition permits different coefficient signs without an algebraic contradiction. It does not establish the empirical cause of a difference. An empty qualifying subset has an undefined conditional mean, whereas its all-post contribution is zero.

The fixed score files and saved daily index are available, but the internal post-ID/date crosswalk required to reaggregate thresholded daily subsets is not distributed. The supplied aggregation code preserves missing conditional means, reports common-denominator contributions at four relevance thresholds and fails explicitly if the crosswalk is absent. No irrelevant-post placebo is treated as newly reproduced support for specificity. The completed mean-relevance, misinformation-score, positive-content and volume-adjusted regressions are reported in Table 27; these remain internal measurement checks, not independent validation of classifier accuracy.

F Heterogeneity and Interaction Checks

F.1 Sector and Listing-Market Sensitivity

The leave-out comparisons retain the fully conditional model and the full sample's tail caps while omitting each energy sub-industry in turn. Table 25 also reports omission of the three largest listing markets. This assesses whether a particular sector or market supplies the fitted association without changing financial tail treatment within every subsample. All use the completed saved estimates. Across sector omissions, coefficients range from approximately -0.0028 to 0.0000. The intervals remain imprecise, and the table does not establish a robust positive coefficient under the fully conditional design. These checks identify the scope of that null result, rather than overwrite the positive association under broader controls. Figure 8 displays the sector comparisons.

F.2 Temporal Interaction

The completed temporal check interacts the exposure index with a post-2019 indicator while retaining security, market-day and sector-month effects and prior-session financial controls. The interaction is -0.0019 with $p = 0.687$. This directly tests a slope difference. It does not show a strengthened spread association after 2019, even though the amount and sector distribution of classified content change. Stored

Table 25: Leave-out diagnostics with fixed full-sample tail caps

Omitted sector or market	Coefficient	SE	p
Coal	-0.0013	0.0031	0.698
Integrated Oil & Gas	-0.0013	0.0026	0.635
Oil & Gas Drilling	-0.0020	0.0022	0.386
Oil & Gas Exploration and Production	-0.0010	0.0026	0.719
Oil & Gas Refining and Marketing	0.0000	0.0027	0.992
Oil & Gas Transportation Services	-0.0015	0.0025	0.572
Oil Related Services and Equipment	-0.0025	0.0021	0.266
Renewable Energy Equipment & Services	-0.0028	0.0025	0.296
Renewable Fuels	-0.0006	0.0021	0.797
Market: Mainland China	-0.0023	0.0032	0.482
Market: Indonesia	0.0002	0.0010	0.838
Market: United States	-0.0020	0.0027	0.481

All specifications retain the principal controls and fixed effects. Leaving out a sector leaves eight sector clusters. The full principal sample supplies tail caps; they are not recomputed within each leave-out sample.

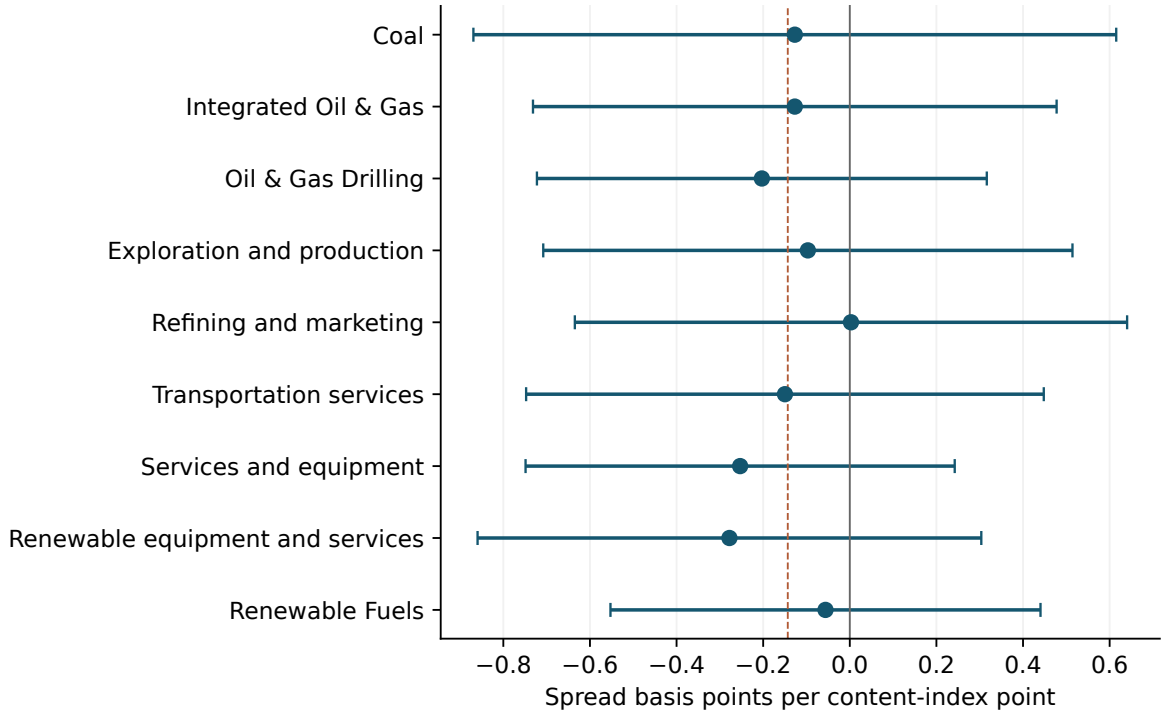


Figure 8: Exposure coefficients after omitting one sector at a time, using the fully conditional model and fixed full-sample tail caps. Bars are approximate 95% sector/date-clustered intervals, with seven reference degrees of freedom.

region-group interactions are not interpreted as language-market effects because the input groups do not map cleanly to actual listing markets.

G Additional Specification and Inference Checks

Table 26: Alternative inference for the principal coefficient

Reference calculation	SE	p
Security/date clusters	0.00108	0.188
Sector/date clusters	0.00220	0.533
Date-score HAC, bandwidth 5	0.00225	0.524
Date-score HAC, bandwidth 20	0.00223	0.521
Date-score HAC, bandwidth 60	0.00201	0.476
Sector score signs	—	0.512

The coefficient and observations are identical in every row. Sector/date clustering has nine sectors. HAC uses ordered observed dates and a normal reference. Score signs enumerate 512 combinations and require independent symmetric sector scores; this is a diagnostic, not exact randomization inference.

Table 27: Alternative content measures and attention adjustment

Content regressor	Coefficient	SE	p
Mean composite	-0.0014	0.0022	0.533
Log of one plus composite sum	0.0002	0.0010	0.820
Mean misinformation score	0.0010	0.0013	0.470
Mean relevance score	-0.0002	0.0004	0.644
Share with misinformation at least 25	0.0338	0.0897	0.716
Positive composite indicator	-0.0036	0.0040	0.398
Composite with volume and relevance controls	-0.0013	0.0023	0.591

Principal controls and fixed effects throughout. Regressor units differ across rows; raw coefficient sizes are not comparable. The final row adds log one plus post count and mean relevance jointly. Sector/date inference.

Table 28: Outcome, sample, and timing sensitivity

Specification	Coefficient	SE	p	N
No financial controls	-0.0012	0.0017	0.492	485,800
Unwinsorized spread and controls	-0.0086	0.0048	0.109	485,800
Log spread	0.0035	0.0020	0.122	485,800
Matched-post dates only	-0.0014	0.0022	0.556	485,332
Previous-session content	-0.0010	0.0022	0.652	485,759
VWAP-midpoint percentage gap	0.0010	0.0033	0.760	485,798

Unless stated otherwise the outcome is a percentage-point spread. Log spread and the VWAP gap have different units/interpretations. No gap outcome is interpreted as transaction price impact. Sector/date inference.

The nine-sector structure motivates comparing sector/date clustering with date-score HAC and the sector score-sign diagnostic. Their assumptions and finite-sample limits are set out in the supplementary material. The full set of completed coefficient estimates is retained, including numerical warnings. In the attention-adjusted model the post-volume control has an undefined sector/date standard error; the exposure coefficient and all coefficients in the fully conditional model have finite reported uncertainty. The absence of a robust positive coefficient in the saturated models limits an independent daily misinformation interpretation without erasing the positive broader association documented in the main text.

Table 29: Financial control definitions and tail caps

Variable	Lower cap	Upper cap
Quoted spread, percent	0.0085	7.5381
Prior log one plus volume	7.9014	19.3589
Prior log market capitalization	20.1426	31.9419
Prior range as percent of close	0.0000	14.7036

Principal sample first and ninety-ninth percentile caps. Exposure scores are not capped.

H Summary

The appendix consolidates the descriptive and diagnostic evidence behind the paper's design. The corpus tables document the scale, heterogeneity, and engagement structure of the Facebook data; the network section maps the dissemination structure of the most active post owners. Together they document how the explanatory variable is constructed from variation in post volume, engagement, and timing.

The financial and exposure sections document coverage, composition and the construction of the exposure index: its distributional properties, the rubric-development spot-check, and the distinction between content availability and observed investor attention. The conditional-mean decomposition explains why relevance subsets need not reproduce the all-post association, while the empirical checks distinguish market-day conditions from sector-month conditions. Read together, the appendix connects the competing-liquidity framework in Section 3 to the positive associations and their empirical boundaries in Section 6.